MIL-M-38510/176A
30 April 1984
SUPERSEDING
MIL-M-38510/176
7 March 1979

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, LATCHES, MONOLITHIC SILICON, POSITIVE LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic, silicon, CMOS logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	8-bit addressable latch
02	Dual 4-bit latch

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
J	D-3 (24-lead, 1/2" x 1-1/4), dual-in-line package
K	F-6 (24-lead, 3/8" x 5/8"), flat package
T	F-8 (24-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm) max
U	F-5 (16-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm) max
Z	F-8 (24-lead, 1/4" x 3/8"), flat package

NOTES:
1. As an exception to nickel plate or undercoating paragraph of MIL-M-38510, for case outlines T and U only, the leads of bottom brazed ceramic packages (i.e., configuration 2 of case outlines F-5 or F-8) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e., finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.
2. For bottom or side brazed packages, case outlines T and U only, the S_1 dimension may go to .000 inch (.00 mm) mimimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: George C, Marshall Space Flight Center, National Aeronautics and Space Administration, ATTN: EG02, Marshall Space Flight Center, AL 35812 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

1.3 Absolute maximum ratings.

Supply voltage range (V_{DD}-V_{SS})	-0.5 Vdc to +18 Vdc
Input current (each input)	±10 mA
Input voltage range	(V_{SS} -0.5) ≤ V_{IN} ≤ (V_{DD} +0.5)
Storage temperature range	-65°C to +175°C
Maximum power dissipation (P_D)	200 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case	(See MIL-M-38510, appendix C)
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Supply voltage range (V_{DD}-V_{SS})	4.5 Vdc to 15 Vdc
Low-level input voltage range (V_{IL})	0-1.5 V dc @ V_{DD} = 5 Vdc, V_{OL} = 10% V_{DD}, V_{OH} = 90 % V_{DD}, 0-4.0 Vdc @ V_{DD} = 15 Vdc, 0-2.0 Vdc @ V_{DD} = 10 Vdc
High level input voltage range (V_{IH})	3.5-5.0 Vdc @ V_{DD} = 5 Vdc, V_{OL} = 10% V_{DD}, V_{OL} = 90% V_{DD}, 11.0-15.0 Vdc @ V_{DD} = 15 V dc, 8.0-10.0 Vdc @ V_{DD} = 10 V dc
Ambient operating temperature range (T_A)	-55°C to +125°C
Load capacitance	50 pF maximum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent, which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Document Improv

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.2.3 Truth tables. The truth tables and logic equations shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1. Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and as specified herein (see 4.5.5).

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 38 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 4, or equivalent. Ambient temperature shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 5, or equivalent. Ambient temperature shall be 125°C minimum. Test duration shall be in accordance with table I of method 1015.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II the failures shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_i measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroup 12 shall be added to the group A inspection requirements for class S devices using an LTPD of 15 and consist of the procedures, test conditions, and limits specified in table III.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 5, or equivalent, shall be used.

b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and shall be measured only for initial qualification and after process or design changes. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Units
Positive clamping Input to V_{DD}	$V_{IC(pos)}$	V_{DD} = GND, V_{SS} = open, output = open, T_A = 25°C, I_{IN} = 1 mA	All	-	1.5	Vdc
Negative clamping input to V_{SS}	$V_{IC(neg)}$	V_{DD} = open, V_{SS} = GND, output = open, T_A = 25°C, I_{IN} =-1 mA	All	-	-6.0	Vdc
Quiescent supply current	I_{SS} or I_{DD}	V_{DD} = 18.0 Vdc, V_{IN} = V_{SS} or V_{DD}, All inputs combinations	All	-	-2.5	µA
High level output voltage	V_{OH1}	V_{DD} = 15.0 Vdc, see table III $\lvert I_O \rvert \leq$ 1 µA	All	14.95	-	Vdc
Low level output voltage	V_{OL1}	V_{DD} = 15.0 Vdc, see table III $\lvert I_O \rvert \leq$ 1 µA	All	-	0.05	Vdc
3-state output leakage current	I_{OC1}	V_{DD} = 18.0 Vdc, each output separately, V_O = V_{DD}	02	-	90	nA
3-state output leakage current	I_{OC2}	V_{DD} = 18.0 Vdc, each output separately, V_O = V_{SS}	02	-	-90	nA
Input high voltage	V_{IH1}	V_{DD} = 5.0 Vdc, see table III $\lvert I_O \rvert \leq$ 1 µA	All	3.5	-	Vdc
	V_{IH2}	V_{DD} = 10.0 Vdc, see table III $\lvert I_O \rvert \leq$ 1 µA	All	7.0	-	Vdc
	V_{IH3}	V_{DD} = 15.0 Vdc, see table III $\lvert I_O \rvert \leq$ 1 µA	All	11.0	-	Vdc
Input low voltage	V_{IL1}	V_{DD} = 5.0 Vdc, see table III $\lvert I_O \rvert \leq$ 1 µA	All	-	1.5	Vdc
	V_{IL2}	V_{DD} = 10.0 Vdc, see table III $\lvert I_O \rvert \leq$ 1 µA	All	-	3.0	Vdc
	V_{IL3}	V_{DD} = 15.0 Vdc, see table III $\lvert I_O \rvert \leq$ 1 µA	All	-	4.0	Vdc
Output high (source) current	I_{OH1}	V_{DD} = 5.0 Vdc, V_{OL} = 4.6 Vdc	-	-	-	mAdc
		Decoder inputs = V_{SS}, All others = V_{DD}	01	-0.36	-	
		Data inputs and strobe = V_{DD} All others = V_{SS}	02	-0.36	-	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Units
Output high (source) current	I_{OH2}	V_{DD} = 15.0 Vdc, V_{OL} = 13.5 Vdc	-	-	-	mAdc
		Decoder inputs = V_{SS}, All others = V_{DD}	01	-2.4	-	
		Data inputs and strobe = V_{DD} All others = V_{SS}	02	-2.4	-	
Output low (sink) current	I_{OL1}	V_{DD} = 5.0 Vdc, V_{OL} = 0.4 Vdc	-	-	-	mAdc
		Decoder inputs and reset = V_{SS}, All others = V_{DD}.	01	0.36	-	
		Reset = V_{DD}, All others = V_{SS}	02	0.36	-	
	I_{OL2}	V_{DD} = 15.0 Vdc V_{OL} = 1.5 Vdc	-	-	-	mAdc
		Decoder inputs and reset = V_{SS}, All others = V_{DD}	01	2.4	-	
		Reset = V_{DD}, All others = V_{SS}	02	2.4	-	
Input leakage current	I_{IH}	V_{DD} = 18.0 Vdc, Measure inputs sequentially; connect all unused inputs to V_{SS}	All	-	45	nA
	I_{IL}	V_{DD} = 18.0 Vdc, Measure inputs sequentially; connect all unused inputs to V_{DD}	All	-	-45	nA
Input capacitance	C_{IN}	V_{DD} = 0 Vdc, f = 1 MHz, T_A = 25°C	All	-	7.5	pF
Input test voltage	V_{ZAP}	C_i = 100 pF, R_z = 1.5 kΩ, (see 4.5.3)	All	400	-	Vdc
Propagation delay times data to output		V_{DD} = 5.0 Vdc, C_L = 50 pF; R_L = 200 kΩ (see figure 6)	01	20	560	ns
High to low level	t_{PHL1}					
Low to high level	t_{PLH1}					
Propagation delay times Write disable to output		V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	01	20	560	ns
High to low level	t_{PHL2}					
Low to high level	t_{PLH2}					

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Units
Propagation delay times Reset to output High to low level	t_{PHL3}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	01	18	490	ns
Address to output High to low level Low to high level	t_{PHL4} t_{PLH4}			23	630	
Transition times High to low level Low to high level	t_{THL} t_{TLH}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	All	10	280	ns
Propagation delay times Strobe to data out High to low level Low to high level	t_{PHL1} t_{PLH1}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	02	17	400	ns
Propagation delay times data in to data out High to low level Low to high level	t_{PHL2} t_{PLH2}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	02	14	400	ns
Propagation delay times reset to data out High to low level Low to high level	t_{PHL3} t_{PLH3}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	02	13	378	ns
3-state output propagation disable delay times Output high to high impedance	t_{PHZ}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	02	11	310	ns
Output low to high impedance	t_{PLZ}		02	9	255	
High impedance to output low	t_{PZL}		02	9	255	
High impedance to output high	t_{PZH}		02	10	275	
Minimum reset pulse width	t_{PR}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	02	-	280	ns
Minimum strobe pulse width	t_{PST}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	02	-	200	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Units
Minimum data setup time Low to high level	t_{SLH}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	02	-	70	ns
Minimum data hold time High to low level	t_{HHL}	V_{DD} = 5.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6)	02	-	0	ns
Propagation delay times data to output High to low level Low to high level	t_{PHL1} t_{PLH1}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	01	7	150	ns
Propagation delay times write disable to output High to low level Low to high level	t_{PHL2} t_{PLH2}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	01	8	160	ns
Propagation delay times reset to output High to low level	t_{PHL3}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	01	8	160	ns
Address to output High to low level Low to high level	t_{PHL4} t_{PLH4}			10	200	
Transition times High to low level Low to high level	t_{THL} t_{TLH}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	All	5	100	ns
Propagation delay times strobe to data out High to low level Low to high level	t_{PHL1} t_{PLH1}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	02	7	140	ns
Propagation delay times data in to data out High to low level Low to high level	t_{PHL2} t_{PLH2}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	02	6	120	ns

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Units
Propagation delay times reset to data out High to low level Low to high level	t_{PHL3} t_{PLH3}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	02	5	100	ns
3-state output propagation disable delay times Output high to high impedance	t_{PHZ}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	02	6	120	ns
Output low to high impedance	t_{PLZ}		02	5	100	
High impedance to output low	t_{PZL}		02	5	100	
High impedance to output high	t_{PZH}		02	5	100	
Minimum reset pulse width	t_{WR}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	02	-	140	ns
Minimum strobe pulse width	t_{WST}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	02	-	80	ns
Minimum data setup time Low to high level	t_{SLH}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	02	-	30	ns
Minimum data hold time High to low level	t_{HHL}	V_{DD} = 10.0 Vdc, C_L = 50 pF R_L = 200 kΩ (see figure 6) T_A = 25°C	02	-	0	ns

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test methods	Class S device 1/				Class B devices 1/			
		Ref. par.	Table III sub-groups 2/	Table IV delta limits 3/	Test circuit figure	Ref. par.	Table III sub-groups 2/	Table IV delta limits 3/	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.2b 4.5.2			4				
3	Same as line 1		1	Δ					
4	Static burn-in II (method 1015)	4.2b 4.5.2			4	4.2b 4.5.2	4/		4
5	Same as line 1		1*	Δ		4.2d	1*	Δ	
6	Dynamic burn-in (method 1015)	4.2b 4.5.2			5				
7	Same as line 1	4.2d	1*	Δ					
8	Final electrical parameters (method 5004)		1*,2,3, 7,9				1*,2,3, 7,9		
9	Group A test requirements (method 5005)	4.4.1	1,2,3,4, 7,8,9, 10,11,12			4.4.1	1,2,3, 4,7,9		
10	Group B end-point electrical parameters (method 5005)	4.4.2	1,2,3,7, 8,9,10, 11	Δ					
11	Group C end-point electrical parameters (method 5005)					4.4.3	1,2,3	Δ	
12	Additional group C tests (method 5005)					4.4.3c	10,11		
13	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3			4.4.4	1,2,3		

1/ Blank spaces indicate tests are not applicable.

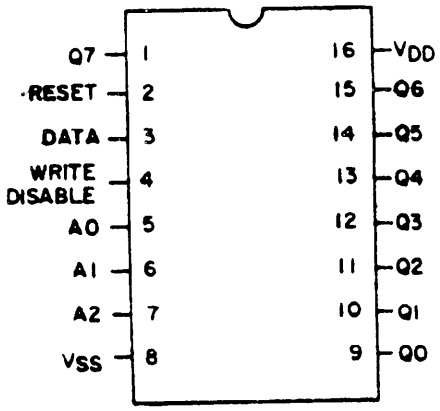
2/ (*) indicates PDA applied to subgroup 1 (see 4.2.1).

3/ (Δ) indicates delta limit shall be required only in table III subgroup 1 where specified, and delta values shall be computed with reference to the previous interim electrical measurements, line 1.

4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 01
Cases E, F, and U



Device type 02
Cases J, K, T, and Z

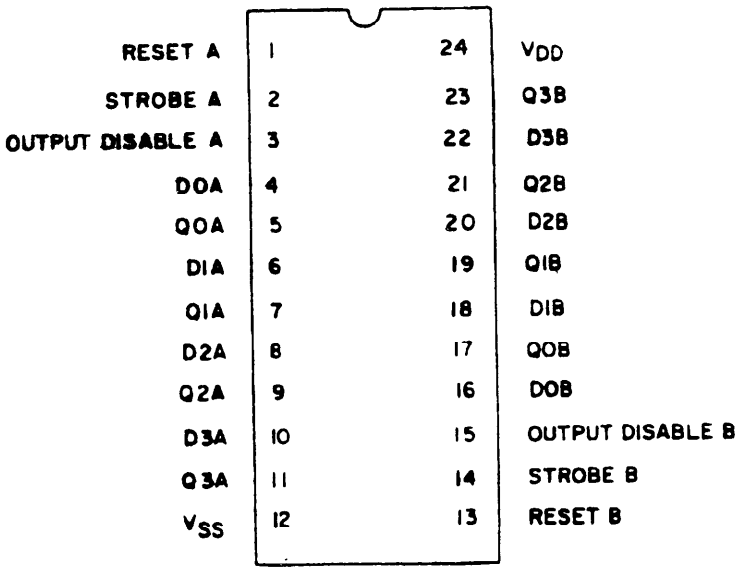
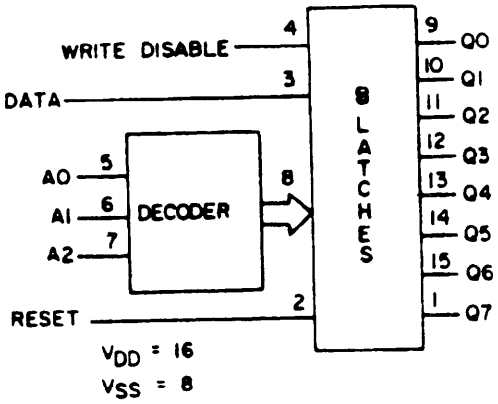
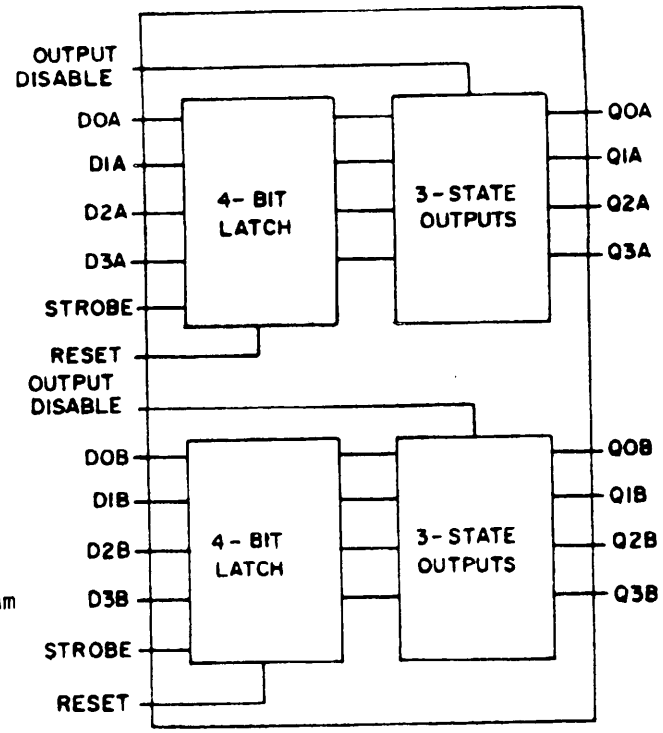


FIGURE 1. Terminal connections.

Device type 01
Cases E, F, and U



Device type 02
Cases J, K, T, and Z



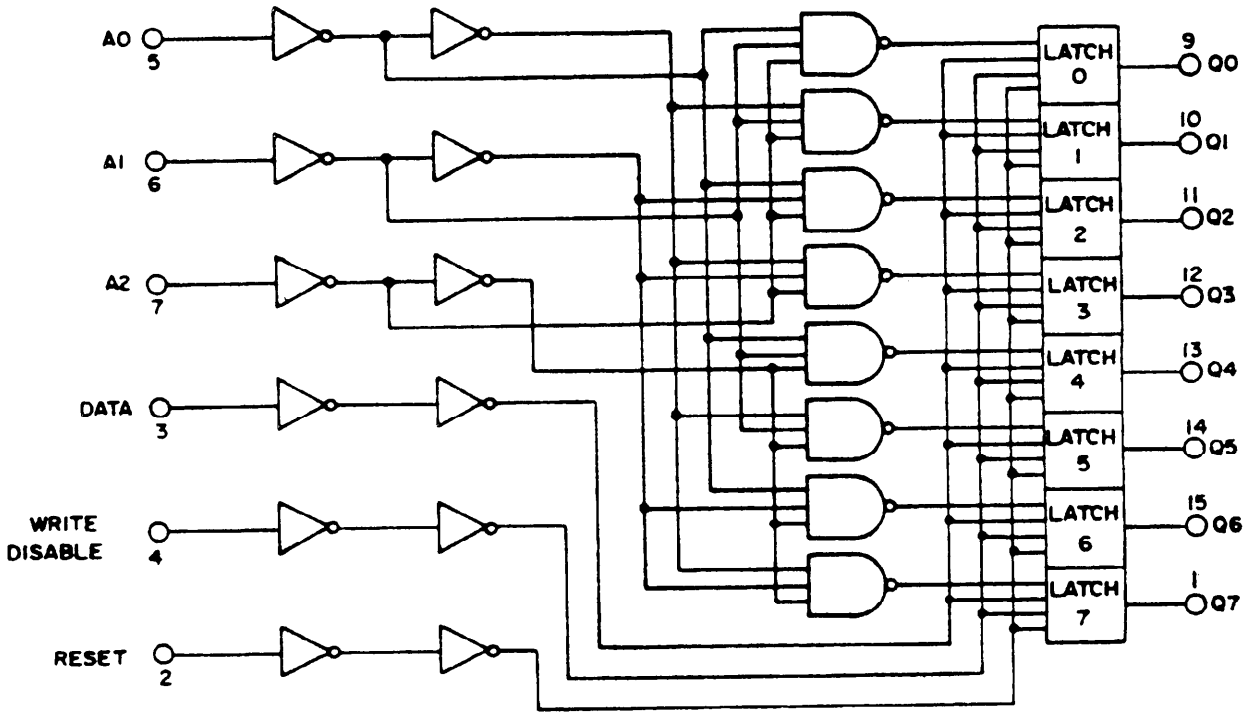
Functional diagram

FIGURE 2. Logic diagrams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 01

Cases E, F, and U



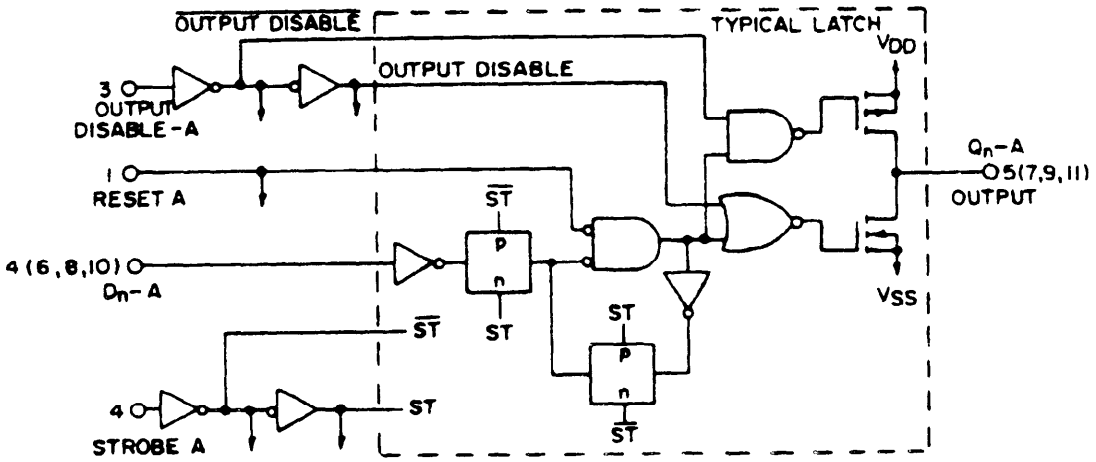
V_{DD} = Pin #16
V_{SS} = Pin #8

FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 02

Cases J, K, T, and Z



Logic diagram

(1 of 4 identical latches)

FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 01

Truth table			
Mode selection			
Inputs		Outputs	
WD	R	Addressed latch	Unaddressed latch
L	L	Follows data	Holds previous state
L	H	Follows data (Active high 8-channel demultiplexer)	Reset to "L"
H	L	Holds previous state	
H	H	Reset to L	Reset to L

WD = Write disable
R = Reset

L = Low level
H = High level

Device type 01

Truth table						
Output pin selection						
R	D	W_D	A_0	A_1	A_2	OUTPUT
L	H	L	L	L	L	Q_0
L	H	L	H	L	L	Q_1
L	H	L	L	H	L	Q_2
L	H	L	H	H	L	Q_3
L	H	L	L	L	H	Q_4
L	H	L	H	L	H	Q_5
L	H	L	L	H	H	Q_6
L	H	L	H	H	H	Q_7

W_D = Write disable
R = Reset
D = Data
A_0 - A_2 = Address select
Q_0 - Q_7 = Output pin
L = Low level
H = High level

Device type 02

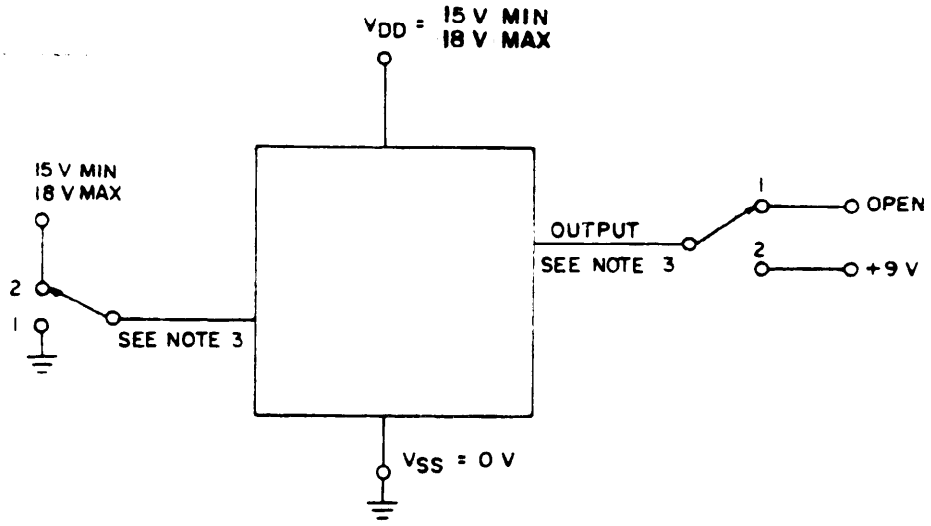
Truth table				
Inputs				Output
Reset	Disable	Strobe	D Input	Q Output
L	L	H	H	H
L	L	H	L	L
L	L	L	X	Latched
H	L	X	X	L
X	H	X	X	Z

H = High level
L = Low level

X = Don't care
Z = High impedance

FIGURE 3. Truth tables.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

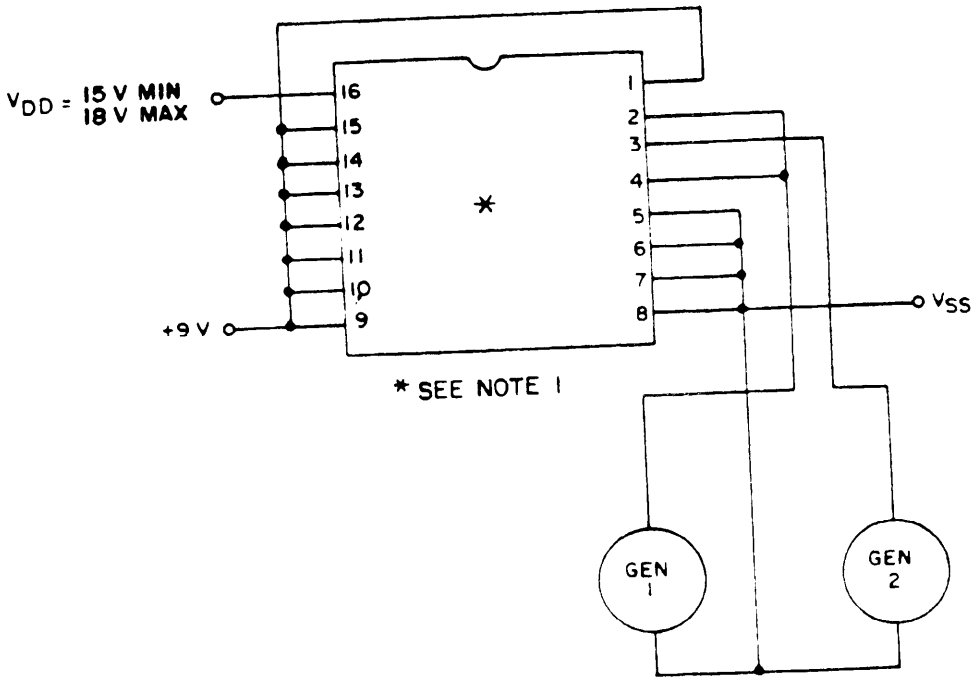


NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected directly to V_{DD} or V_{SS} or through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.

FIGURE 4. Static burn-in test circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 01

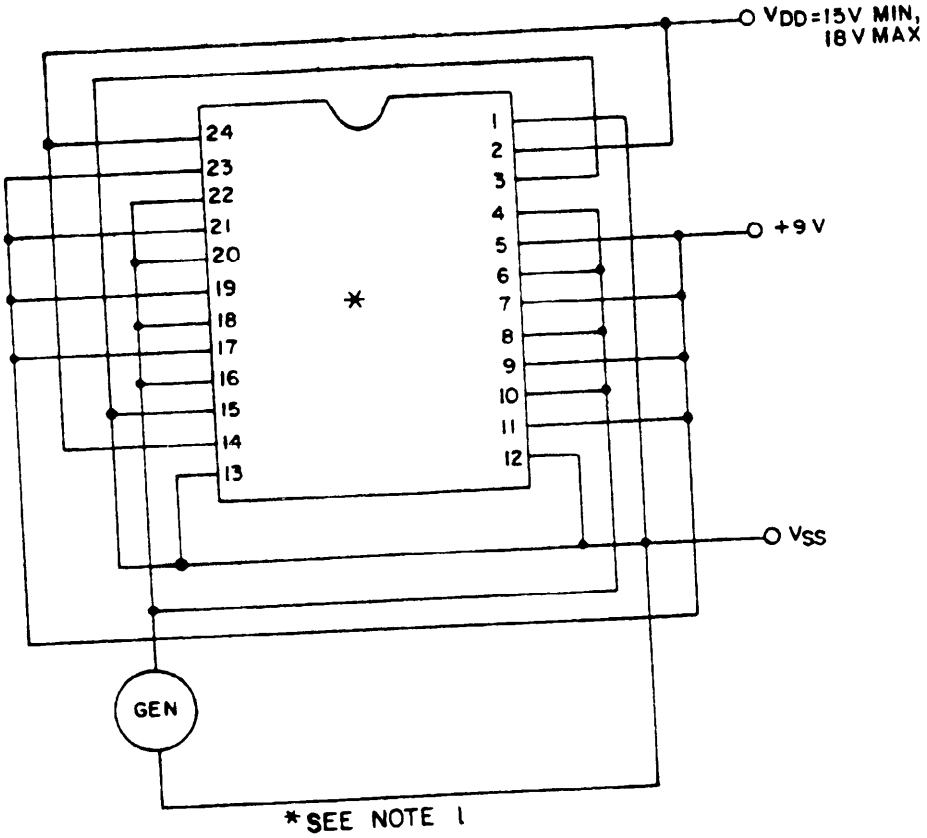


NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < 1 MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5, -10% V_{DD}.

FIGURE 5. Dynamic burn-in and steady-state life test circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 02



NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < 1 MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5, -10% V_{DD}.

FIGURE 5. Dynamic burn-in and steady state life test - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 01

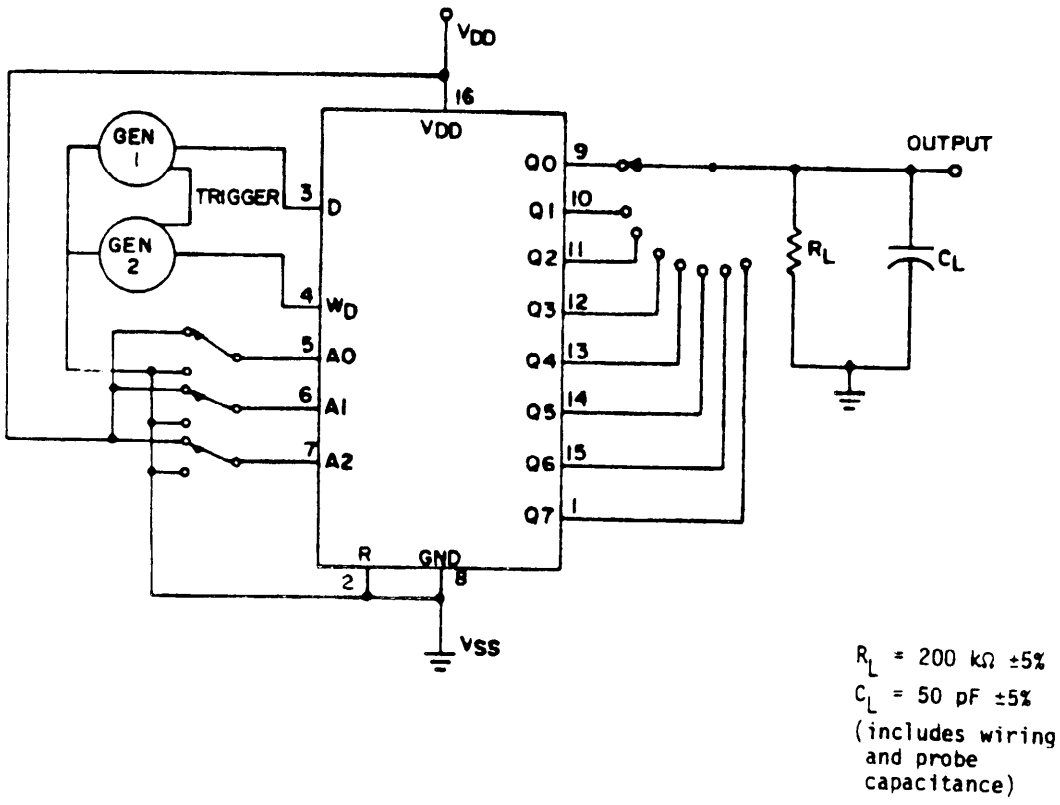
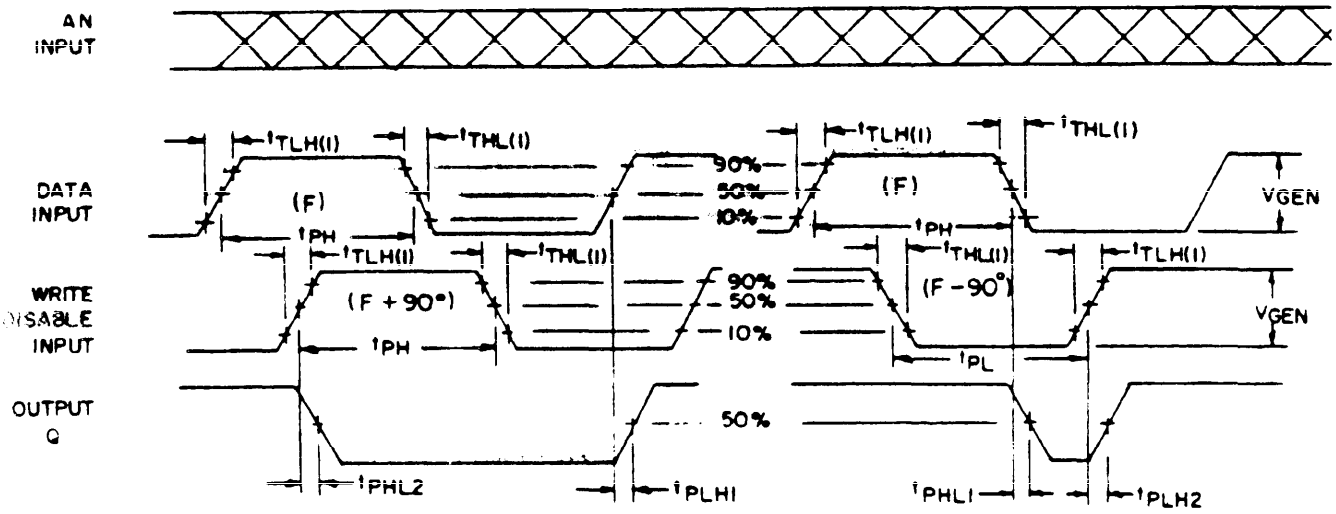


FIGURE 6. Switching time test circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

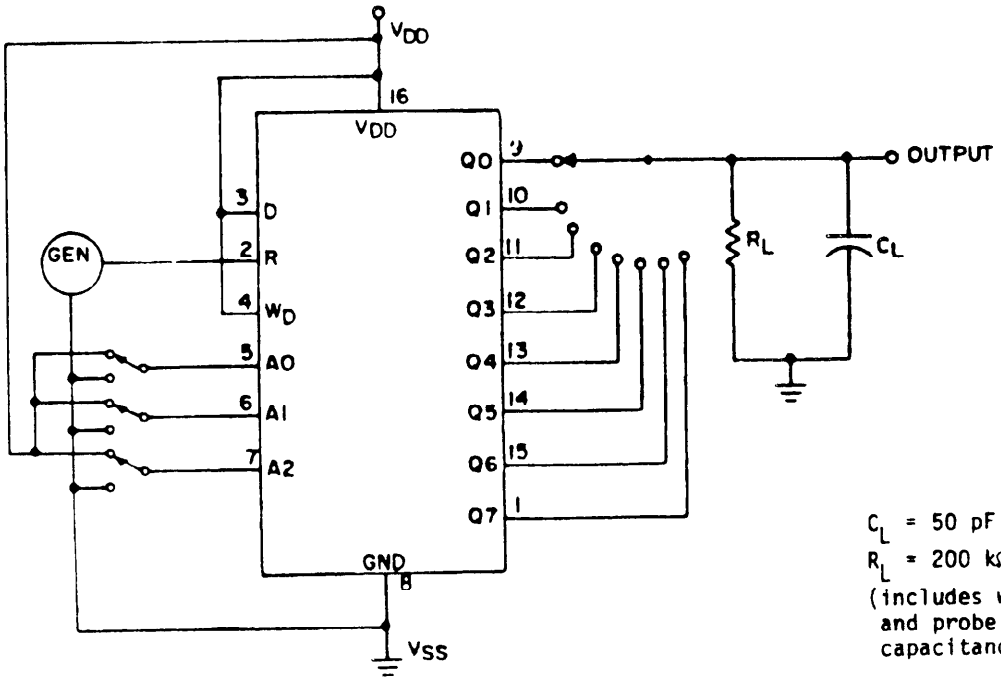


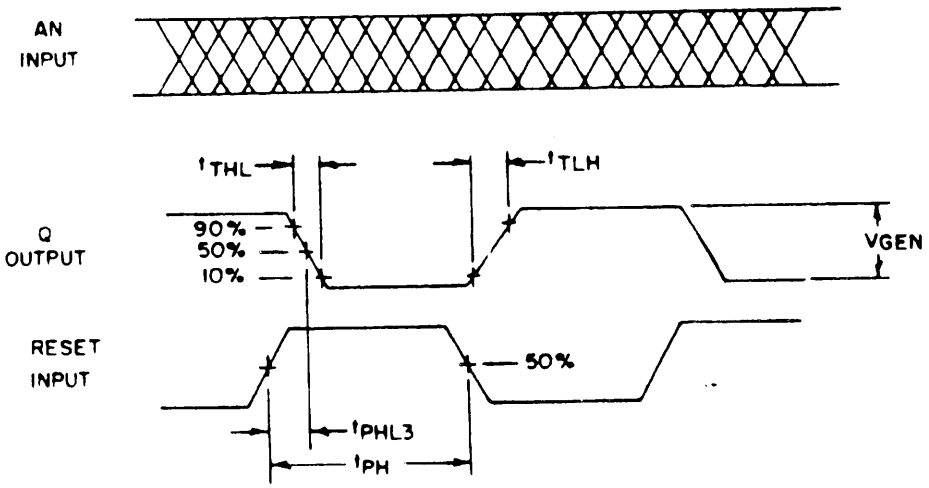
NOTES:
1. The pulse generator number 1 has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0 \pm 0.1$ us, $t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns and PRR = 100 kHz.
2. The pulse generator number 2 has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0 \pm 0.1$ us, $t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns, and PRR = 50 kHz.
3. See table III for complete terminal conditions.

FIGURE 6. Switching time test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 01



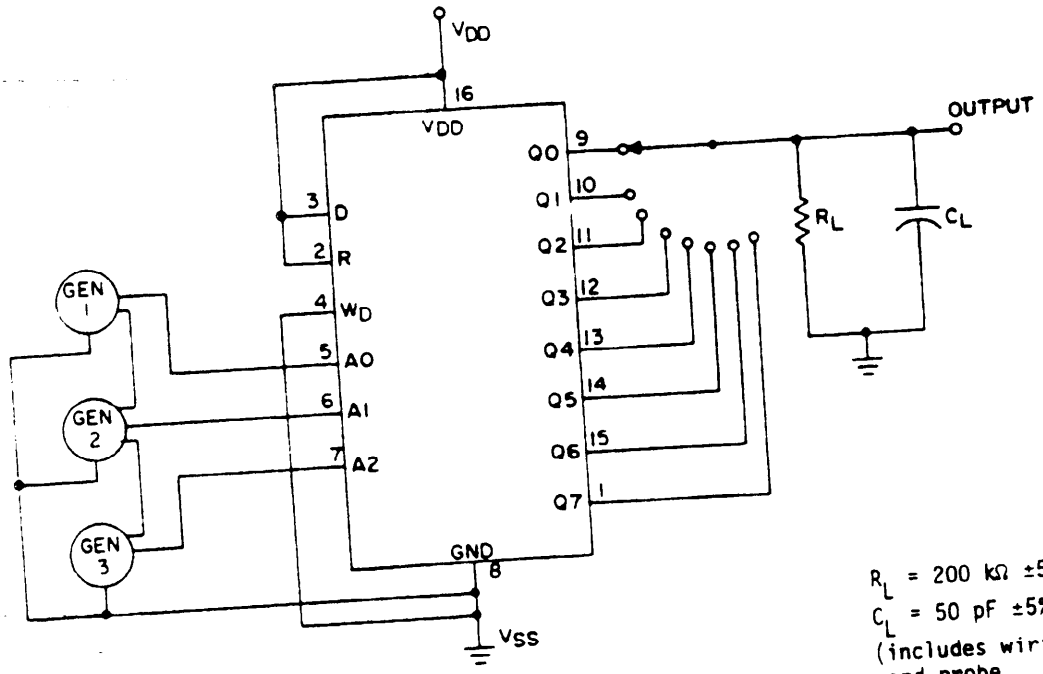


NOTES:
1. The pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns and PRR = 100 kHz.
2. See table III for complete terminal conditions.

FIGURE 6. Switching time test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 01

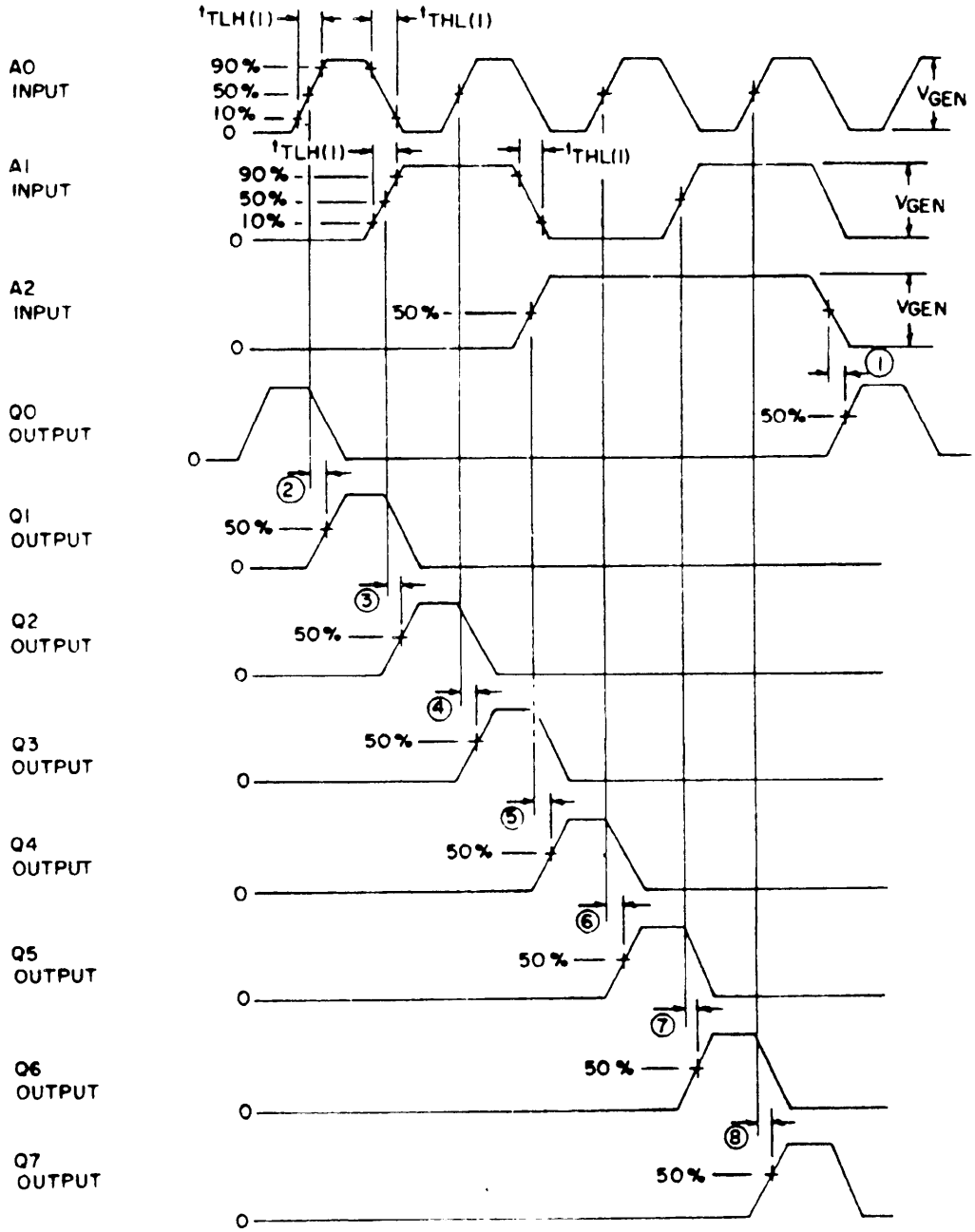


NOTES:
1. The pulse generator number 1 has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns and PRR = 100 kHz.
2. The pulse generator number 2 has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns and PRR = 50 kHz.
3. The pulse generator number 3 has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns and PRR = 25 kHz.
4. See table III for complete terminal conditions.

FIGURE 6. Switching time test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 01



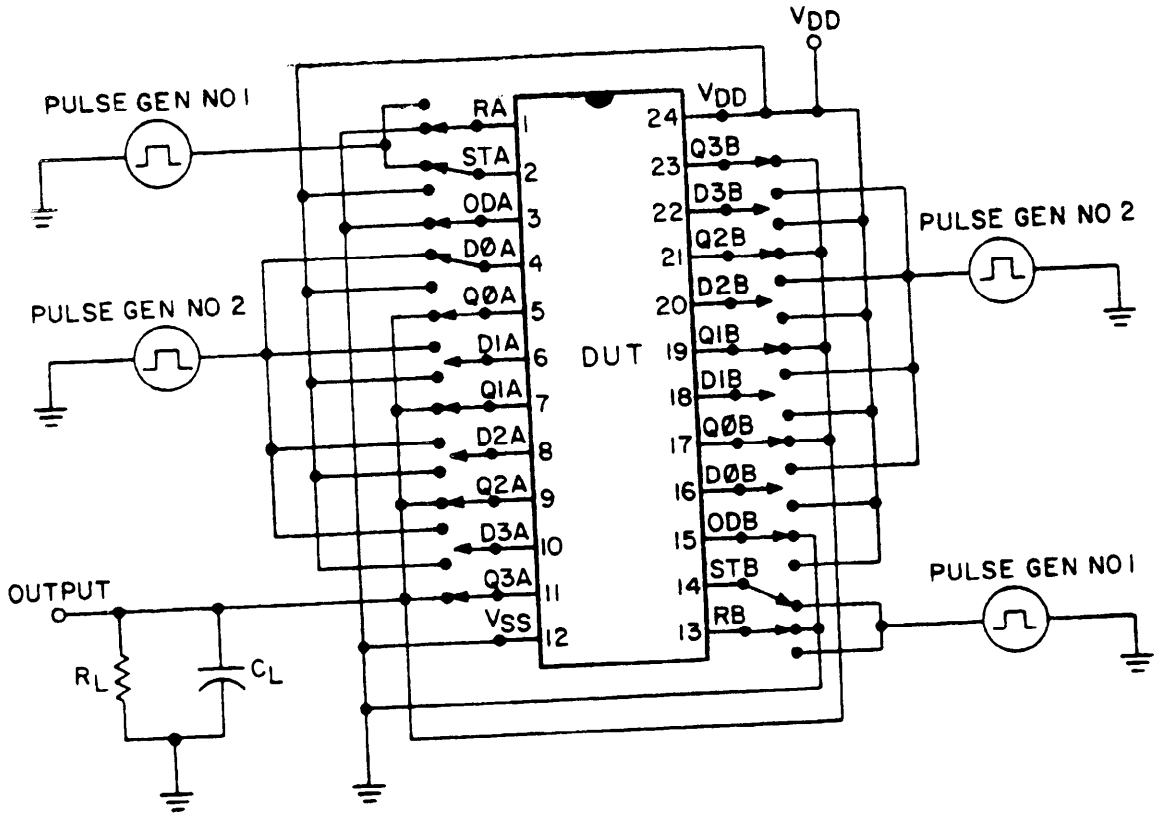
t_{PLH4}		
	A	to Q
①	2	0
②	0	1
③	1	2
④	0	3
⑤	2	4
⑥	0	5
⑦	1	6
⑧	0	7

Example: Above table t_{PLH4} for A2 to Q0 is indicated by ① on the timing diagram.

FIGURE 6. Switching time test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 02



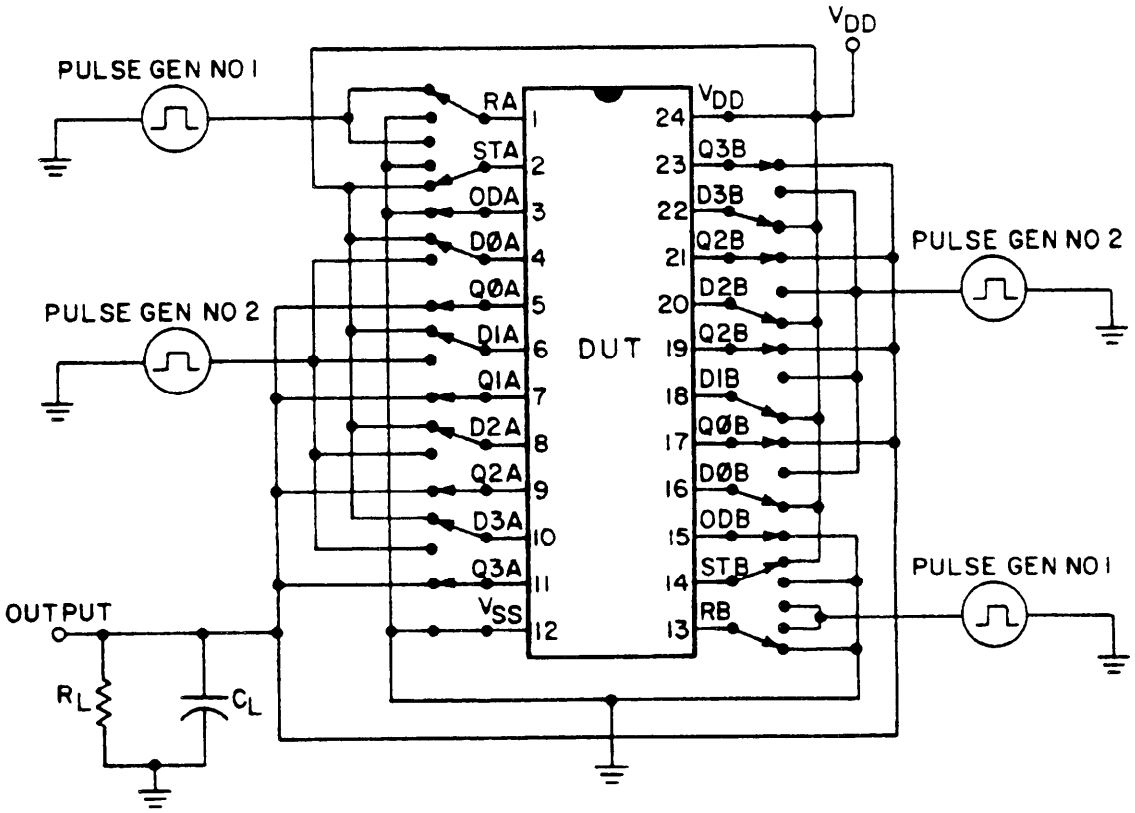
R_L = 200 kΩ ±5%.
C_L = 50 pF ±5%
(Includes wiring and probe capacitance).

NOTES:
1. See notes for input pulse conditions and waveforms. This test circuit applies only to t_{THL}, t_{TLH}, t_{PHL1}, t_{PLH1}, t_{PHL2} t_{PLH2}, t_{PHL3}, and t_{PLH3}.

FIGURE 6. Switching time test circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 02

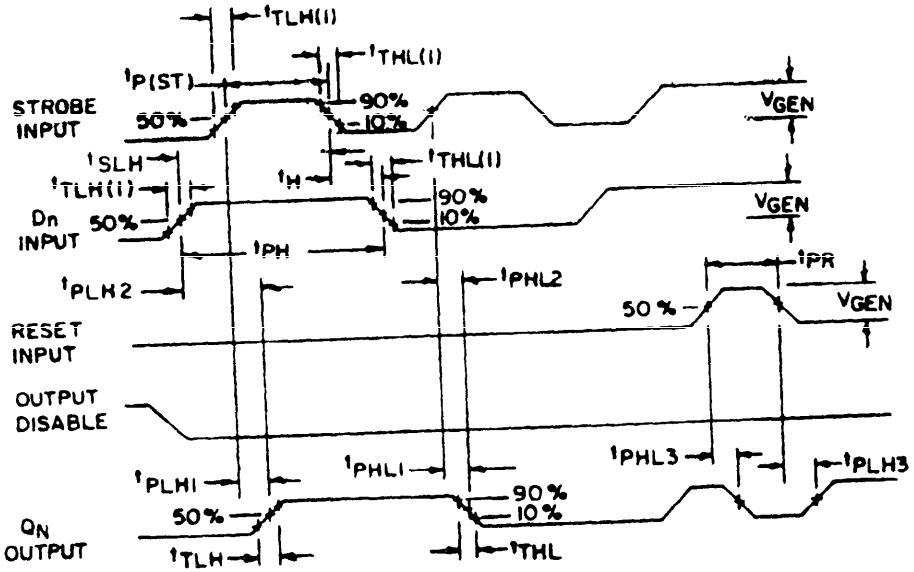


R_L = 200 kΩ ±5%.
C_L = 50 pF ±5%
(Includes wiring and probe capacitance).

NOTES:
1. See notes for input pulse conditions and waveforms.
2. This test circuit applies only to t_{PR}, t_{PS}, t_{SLH}, and t_{HHL}.

FIGURE 6. Switching time test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

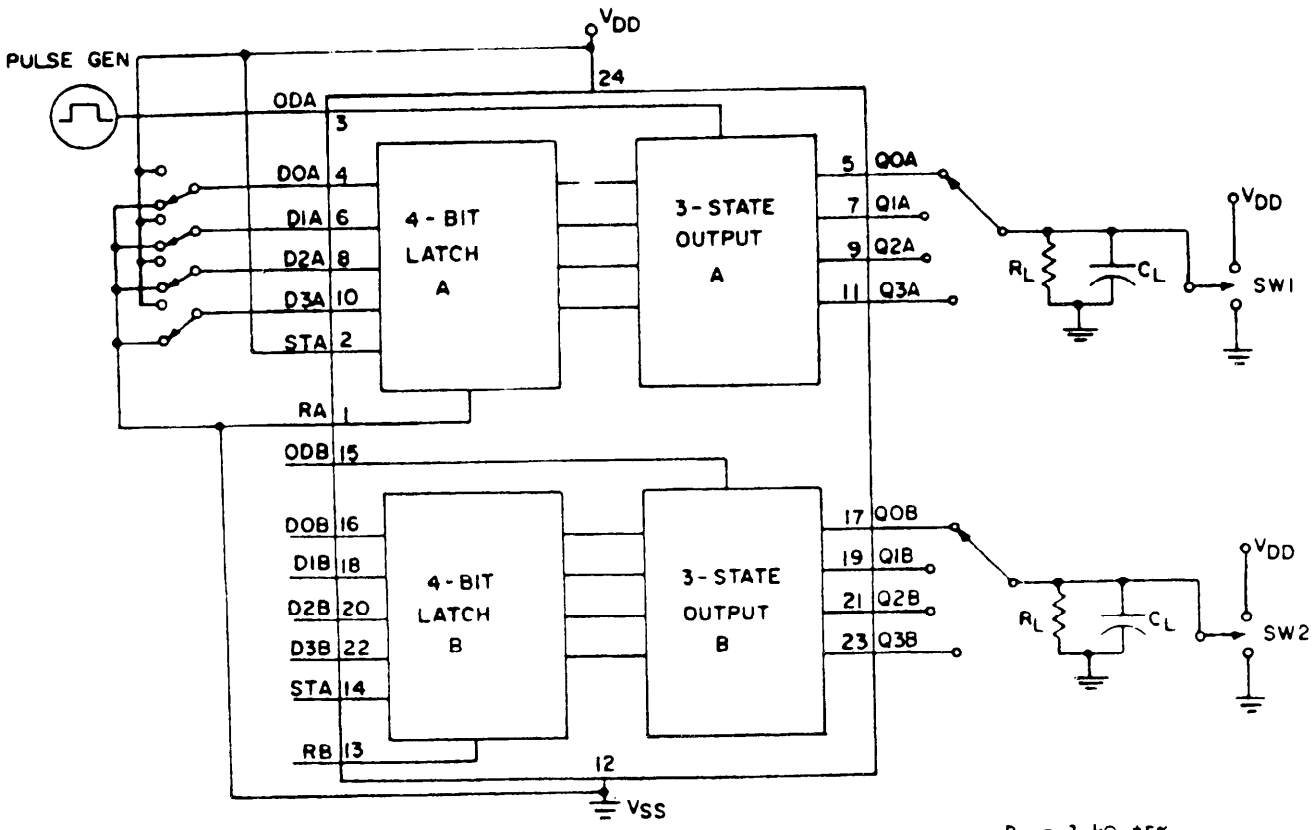


NOTES:
1. The pulse generator number 1 has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns and PRR = 200 kHz.
2. The pulse generator number 2 has the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns and PRR = 100 kHz.
3. See table III for complete terminal conditions.
4. Identical measurements shall be performed on latch A and latch B.
5. Requirements for t_{PR}, t_{PS}, t_{SLH} and t_{HHL} are established by setting the parameter to the limit given in table III and observing proper output state change.

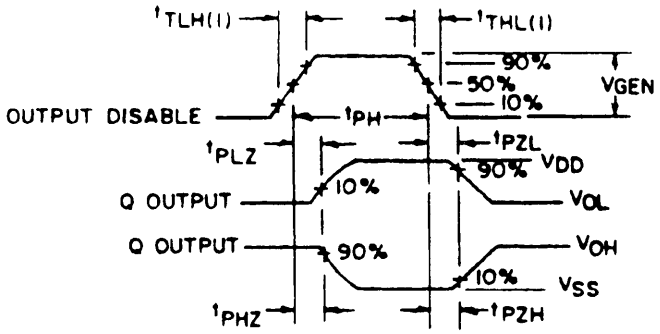
FIGURE 6. Switching time test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Device type 02



R_L = 1 kΩ ±5%
C_L = 50 pF ±5%
(includes wiring and probe capacitance)



NOTES:
1. The pulse generator has the following characteristics: $V_{GEN} = V_{DD}$ ±1.0%, t_{PH} = 1.0 ±0.1 µs, $t_{TLH(1)} = t_{THL(1)}$ = 20 ±2 ns and PRR = 200 kHz.
2. See table III for complete terminal conditions.
3. Identical measurements shall be performed on latch A and latch B.
4. For t_{PHZ} and t_{PZH}, SW1 shall be connected to V_{SS}.
5. For t_{PLZ} and t_{PZL}, SW1 shall be connected to V_{DD}.

FIGURE 6. Switching time test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases E,F,U	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	Q_7	R	D	W_D	A_0	A_1	A_2	V_{SS}	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
V_{IC} (pos)		1		1mA														GND	R		1.5					Vdc
		2			1mA														D							
		3				1mA													W_D							
		4					1mA												A_0							
		5						1mA											A_1							
		6							1mA										A_2							
V_{IC} (neg)		7		-1mA						GND									R		-6.0					
		8			-1mA														D							
		9				-1mA													W_D							
		10					-1mA												A_0							
		11						-1mA											A_1							
		12							-1mA										A_2							
I_{SS} 2/ 3/	3005	13		GND	GND	GND	GND	GND	GND									18.0 V	V_{SS}		-.25		-2.5			µA
		14		18.0 V	GND	GND	GND	GND	GND																	
		15		18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V																	
		16		GND		18.0 V	18.0 V	18.0 V	18.0 V																	
		17		GND		18.0 V	GND	GND	GND																	
		18		GND		GND	GND	GND	GND																	
		19		18.0 V	GND	GND	GND	GND	GND																	
V_{IH1}		20	4/	4/	4/	4/	4/	4/	4/		4/	4/	4/	4/	4/	4/	4/	4/	All outputs		4/		4/		4/	
V_{IH2}		21																								
V_{IH3}		22																								
V_{IL1}		23																								
VIL2		24																								
V_{IL3}		25																								

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,U Test no.	Terminal conditions 1/ $\overline{Q_7}$ 1	R 2	D 3	V_D 4	A_0 5	A_1 6	A_2 7	V_{SS} 8	Q_0 9	Q_1 10	Q_2 11	Q_3 12	Q_4 13	Q_5 14	Q_6 15	V_{DD} 16	Measured terminal	Test limits Subgroup 1 T_A=25°C Min	Max	Subgroup 2 T_A=125°C Min	Max	Subgroup 3 T_A=-55°C Min	Max	Unit
V_{OH1}	3006	26		GND	15 V	GND	GND	GND	GND	GND								15 V	Q_0	14.95		14.95		14.95		Vdc
		27					15 V	GND											Q_1							
		28					GND	15 V											Q_2							
		29					15 V	15 V											Q_3							
		30					GND	GND	15 V										Q_4							
		31					15 V	GND											Q_5							
		32					GND	15 V											Q_6							
		33					15 V	15 V											Q_7							
V_{OL1}	3007	34		15 V			GND	GND	GND										Q_0		0.05		0.05		0.05	Vdc
		35					15 V	GND											Q_1							
		36					GND	15 V											Q_2							
		37					15 V	15 V											Q_3							
		38					GND	GND	15 V										Q_4							
		39					15 V	GND											Q_5							
		40					GND	15 V											Q_6							
		41					15 V	15 V											Q_7							
I_{IL1}		42		GND	5 V	5 V	GND	GND			0.4 V							5.0 V	Q_0	0.51		0.36		0.64		mA
		43										0.4 V							Q_1							
		44											0.4 V						Q_2							
		45												0.4 V					Q_3							
		46													0.4 V				Q_4							
		47														0.4 V			Q_5							
		48															0.4 V		Q_6							
		49	0.4 V																Q_7							

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 8/26/2024 4:13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,U	Terminal conditions 1/																Measured terminal	Test limits						Unit
			O_7	R	D	W_D	A_0	A_1	A_2	V_{SS}	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	V_{DD}		Subgroup 1 T_A=25°C		Subgroup 2 T_A=125°C		Subgroup 3 T_A=-55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
I_{OH1}		50		GND	5 V	GND	GND	GND	GND	GND	4.6 V							5.0 V	Q_0	-0.51		-0.36		-0.64		mA
		51					5 V					4.6 V							Q_1							
		52					GND						4.6 V						Q_2							
		53					5 V							4.6 V					Q_3							
		54					GND	5 V							4.6 V				Q_4							
		55					5 V									4.6 V			Q_5							
		56					GND										4.6 V		Q_6							
		57	4.6 V				5 V												Q_7							
I_{OL2}		58			15 V	15 V	GND	GND			1.5 V							15.0 V	Q_0	3.4		2.4		4.2		
		59										1.5 V							Q_1							
		60											1.5 V						Q_2							
		61												1.5 V					Q_3							
		62													1.5 V				Q_4							
		63														1.5 V			Q_5							
		64															1.5 V		Q_6							
		65	1.5 V																Q_7							
I_{OH2}		66				GND	GND				13.5 V								Q_0	-3.4		-2.4		-4.2		
		67					15 V					13.5 V							Q_1							
		68					GND						13.5 V						Q_2							
		69					15 V							13.5 V					Q_3							
		70					GND	15 V							13.5 V				Q_4							
		71					15 V									13.5 V			Q_5							
		72					GND										13.5 V		Q_6							
		73	13.5 V				15 V												Q_7							
I_{IH1} 5/	3010	74		18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V									18.0 V	All inputs together		6					nA

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,U	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	Q_7	R	D	W_D	A_0	A_1	A_2	V_{SS}	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
I_{IH2}	3010	75		18.0 V	GND	GND	GND	GND	GND	GND							18.0 V		R		1		45			nA
		76		GND	18.0 V	GND	GND												D							
		77			GND	18.0 V	GND												W_D							
		78				GND	18.0 V												A_0							
		79					GND	18.0 V											A_1							
		80					GND	GND	18.0 V										A_2							
I_{IL2} 5/	3009	81					GND	GND	GND										All inputs together		-6					
		82			18.0 V	18.0 V	18.0 V	18.0 V	18.0 V										R		-1		-45			
		83		18.0 V	GND	18.0 V	18.0 V												D							
		84			18.0 V	GND	18.0 V												W_D							
		85				18.0 V	GND												A_0							
		86				18.0 V	18.0 V	GND											A_1							
		87				18.0 V		18.0 V	GND										A_2							
																				Subgroup 4 T_A = 25°C						
																				Min	Max					
C_{IN} 4/	3012	88		6/						GND							GND		R		7.5					pF
		89			6/														D							
		90				6/													W_D							
		91					6/												A_0							
		92						6/											A_1							
		93							6/										A_2							
Truth table test	3014	94	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	7/	All outputs	Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		T_A = -55°C		
																				See 7/ 9/						

See footnotes at end of table.

Source: https://asap.dla.mil -- Downloaded: 2026-10-21T13:45:52Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,U	Terminal conditions 1/																Measured terminal	Test limits						Units
																				Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Symbol	Q_7	R	D	W_D	A_0	A_1	A_2	V_{SS}	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	V_{DD}								
		Test No.																		Min	Max	Min	Max	Min	Max	
t_{PLH1}	3003 Fig. 6	95		GND	IN1	IN2	GND	GND	GND	GND	OUT							5.0 V	D to Q_0	20	400	28	560	20	400	ns
		96				8/	5.0 V	GND				OUT							D to Q_1							
		97					GND	5.0 V					OUT						D to Q_2							
		98					5.0 V	5.0 V						OUT					D to Q_3							
		99					GND	GND	5.0 V						OUT				D to Q_4							
		100					5.0 V	GND								OUT			D to Q_5							
		101					GND	5.0 V									OUT		D to Q_6							
		102	OUT				5.0 V	5.0 V											D to Q_7							
t_{PHL1}		103					GND	GND	GND		OUT								D to Q_0							
		104					5.0 V	GND				OUT							D to Q_1							
		105					GND	5.0 V					OUT						D to Q_2							
		106					5.0 V	5.0 V						OUT					D to Q_3							
		107					GND	GND	5.0 V						OUT				D to Q_4							
		108					5.0 V	GND								OUT			D to Q_5							
		109					GND	5.0 V									OUT		D to Q_6							
		110	OUT				5.0 V	5.0 V											D to Q_7							
t_{PLH2}		111					GND	GND	GND		OUT								W_D to Q_0							
		112					5.0 V	GND				OUT							W_D to Q_1							
		113					GND	5.0 V					OUT						W_D to Q_2							
		114					5.0 V	5.0 V						OUT					W_D to Q_3							
		115					GND	GND	5.0 V						OUT				W_D to Q_4							
		116					5.0 V	GND								OUT			W_D to Q_5							
		117					GND	5.0 V									OUT		W_D to Q_6							
		118	OUT				5.0 V	5.0 V											W_D to Q_7							
t_{PHL2}		119					GND	GND	GND		OUT								W_D to Q_0							
		120					5.0 V	GND				OUT							W_D to Q_1							

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,U	Terminal conditions 1/																Measured terminal	Test limits						Unit
			$\overline{Q}_7$	R	D	$\overline{W}_D$	A_0	A_1	A_2	V_{SS}	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	V_{DD}		Subgroup 9 T_A=25°C		Subgroup 10 T_A=125°C		Subgroup 11 T_A=-55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PHL2}	3003 Fig. 6	121		GND	IN1	IN2 8/	GND	5.0 V	GND	GND			OUT					5.0 V	$\overline{W}_D$ to Q_2	20	400	28	560	20	400	ns
		122					5.0 V	5.0 V	GND					OUT					$\overline{W}_D$ to Q_3							
		123					GND	GND	5.0 V						OUT				$\overline{W}_D$ to Q_4							
		124					5.0 V	GND								OUT			$\overline{W}_D$ to Q_5							
		125					GND	5.0 V									OUT		$\overline{W}_D$ to Q_6							
		126	OUT				5.0 V	5.0 V											$\overline{W}_D$ to Q_7							
t_{PHL3}		127		IN	5.0 V	5.0 V	GND	GND	GND		OUT								R to Q_0	18	350	25	490	18	350	
		128					5.0 V	GND				OUT							R to Q_1							
		129					GND	5.0 V					OUT						R to Q_2							
		130					5.0 V	5.0 V						OUT					R to Q_3							
		131					GND	GND	5.0 V						OUT				R to Q_4							
		132					5.0 V	GND								OUT			R to Q_5							
		133					GND	5.0 V									OUT		R to Q_6							
		134	OUT				5.0 V	5.0 V											R to Q_7							
t_{PLH4} and t_{PHL4}		135		GND		GND	IN1	IN2	IN3		OUT								A_2 to Q_0	23	450	32	630	23	450	
		136										OUT							A_0 to Q_1							
		137											OUT						A_1 to Q_2							
		138												OUT					A_0 to Q_3							
		139													OUT				A_2 to Q_4							
		140														OUT			A_0 to Q_5							
		141															OUT		A_1 to Q_6							
		142	OUT																A_0 to Q_7							
t_{TLH}	3004 Fig. 6	143			IN	8/	GND	GND	GND		OUT								Q_0	10	200	14	280	10	200	
		144					5.0 V	GND				OUT							Q_1	10	200	14	280	10	200	
		145					GND	5.0 V					OUT						Q_2	10	200	14	280	10	200	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-01-13 11:13:45457
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,U	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	Q_7	R	D	W_D	A_0	A_1	A_2	V_{SS}	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	V_{DD}								
		Test No.																		Min	Max	Min	Max	Min	Max	
t_{TLH}	3004 Fig. 6	146		GND	IN	8/	5.0 V	5.0 V	GND	GND				OUT				5.0 V	Q_3	10	200	14	280	10	200	ns
		147					GND	GND	5.0 V						OUT				Q_4							
		148					5.0 V	GND								OUT			Q_5							
		149					GND	5.0 V									OUT		Q_6							
		150	OUT				5.0 V	5.0 V											Q_7							
t_{THL}		151					GND	GND	GND		OUT								Q_0							
		152					5.0 V	GND				OUT							Q_1							
		153					GND	5.0 V					OUT						Q_2							
		154					5.0 V	5.0 V						OUT					Q_3							
		155					GND	GND	5.0 V						OUT				Q_4							
		156					5.0 V	GND								OUT			Q_5							
		157					GND	5.0 V									OUT		Q_6							
		158	OUT				5.0 V	5.0 V											Q_7							
																				Subgroup 12 T_A = 25°C						
t_{PLH1} and t_{PHL1}	3003 Fig. 6	159		GND	IN1	IN2	GND	GND	GND	GND	OUT							10.0 V	D to Q_0	7	150					ns
		160				8/	10 V	GND				OUT							D to Q_1							
		161					GND	10 V					OUT						D to Q_2							
		162					10 V	10 V						OUT					D to Q_3							
		163					GND	GND	10 V						OUT				D to Q_4							
		164					10 V	GND								OUT			D to Q_5							
		165					GND	10 V									OUT		D to Q_6							
		166	OUT				10 V	10 V											D to Q_7							
t_{PLH2} and t_{PHL2}		167					GND	GND	GND		OUT								W_D to Q_0	8	160					
		168					10 V	GND				OUT							W_D to Q_1							
		169					GND	10 V					OUT						W_D to Q_2							

See footnotes at end of table.

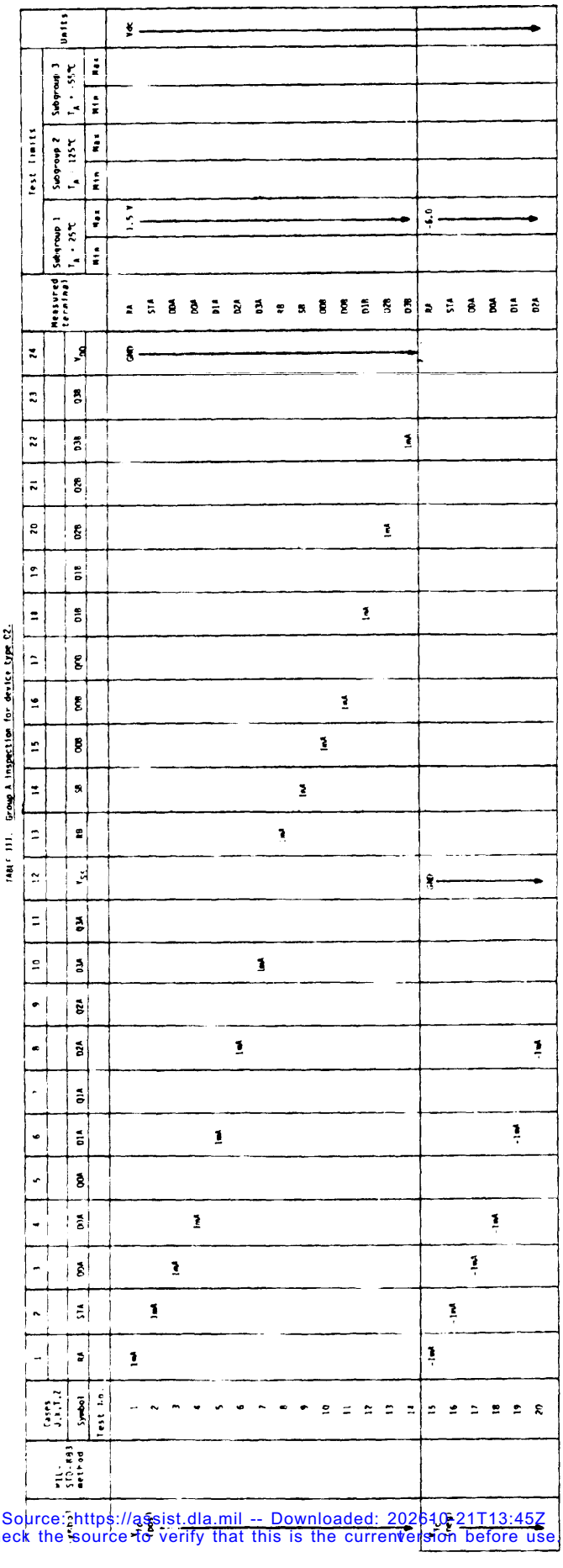
Source: https://assist.dla.mil - Downloaded: 2026-04-21T13:49:47Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,U Test no.	Terminal conditions 1/ Q_7 1	R 2	D 3	W_D 4	A_0 5	A_1 6	A_2 7	V_{SS} 8	Q_0 9	Q_1 10	Q_2 11	Q_3 12	Q_4 13	Q_5 14	Q_6 15	V_{DD 16	Measured terminal	Test limits Subgroup 12 T_A=25°C Min	Max	Min	Max	Min	Max	Unit
t_{PLH2} and t_{PHL2}	3003 Fig. 6	170	OUT	GND	IN1	IN2 8/	10 V	10 V	GND	GND				OUT				10.0 V	W_D to Q_3	8	160					ns
		171					GND	GND	10 V						OUT				W_D to Q_4							
		172					10 V	GND								OUT			W_D to Q_5							
		173					GND	10 V									OUT		W_D to Q_6							
		174					10 V	10 V											W_D to Q_7							
t_{PLH}		175		IN	10 V	10 V	GND	GND	GND		OUT								R to Q_0							
		176					10 V	GND				OUT							R to Q_1							
		177					GND	10 V					OUT						R to Q_2							
		178					10 V	10 V						OUT					R to Q_3							
		179					GND	GND	10 V						OUT				R to Q_4							
		180					10 V	GND								OUT			R to Q_5							
		181					GND	10 V									OUT		R to Q_6							
		182	OUT				10 V	10 V											R to Q_7							
t_{TLH4} and t_{THL4}		183		GND		GND	IN1	IN2	IN3		OUT								A_2 to Q_0	10	200					
		184										OUT							A_0 to Q_1							
		185											OUT						A_1 to Q_2							
		186												OUT					A_0 to Q_3							
		187													OUT				A_2 to Q_4							
		188														OUT			A_0 to Q_5							
		189															OUT		A_1 to Q_6							
		190	OUT																A_0 to Q_7							
t_{TLH} and t_{THL}	3004 Fig. 6	191			IN	8/	GND	GND	GND		OUT								Q_0	5	100					
		192					10 V	GND				OUT							Q_1							
		193					GND	10 V					OUT						Q_2							
		194					10 V	10 V						OUT					Q_3							
		195					GND	GND	10 V						OUT				Q_4							
		196					10 V	GND								OUT			Q_5							
		197					GND	10 V									OUT		Q_6							
		198	OUT				10 V	10 V											Q_7							

Source: https://assist.dla.mil -- Downloaded: 2026-06-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02.



Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

MIL-STD-883 method	Cases 1, 2, 4	Symbol	Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits: Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
		RA		GATE	GND	GND	18.0 V	18.0 V	GND																		GND	D3A							Vdc
		S1A		GND	18.0 V	18.0 V	GND																					R8							
		D0A		GND	18.0 V	18.0 V	GND																					S8							
		Q0A		18.0 V	GND			18.0 V																				D0B							
		D1A		GND	18.0 V				GND	18.0 V																		D1B							
		Q1A		18.0 V	GND				18.0 V																			D2B							
		D2A		GND	18.0 V				GND	18.0 V																		D3B							
		Q2A		18.0 V	GND				18.0 V																			V_{SS}							
		D3A		GND	18.0 V				GND	18.0 V																			Q0A							
		V_{SS}		GND																								Q1A							
		R8																										Q2A							
		S8																										Q3A							
		D0B																										Q0B							
		D1B																										Q1B							
		D2B																										Q2B							
		D3B																										Q3B							
		V_{DD}																										V_{DD}							

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,T,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits: Subgroup 1 T_A = 25°C Min	Subgroup 1 T_A = 25°C Max	Subgroup 2 T_A = 125°C Min	Subgroup 2 T_A = 125°C Max	Subgroup 3 T_A = -55°C Min	Subgroup 3 T_A = -55°C Max	Units
		Symbol	RA	STA	ODA	DOA	QOA	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	SB	OOB	DOB	QOB	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}								
		Test No.																																	
ICC12		41	GND	GND	18.0 V	GND		GND		GND		GND		GND	GND	GND	18.0 V	GND		GND		GND		GND	18.0 V	18.0 V	Q3B		25		90		25	mA
ICC11		42					18.0 V		18.0 V		18.0 V		18.0 V						18.0 V		18.0 V		18.0 V		18.0 V		All outputs together		400					
ICC22		43					GND																				Q0A		-25		-90		-25	
		44							GND																		Q1A							
		45									GND																Q2A							
		46											GND														Q3A							
		47																	GND								Q0B							
		48																			GND						Q1B							
		49																					GND				Q2B							
		50																							GND		Q3B							
ICC21		51					GND		GND		GND		GND						GND		GND		GND		GND		All outputs together		-400					
IIL1		52	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/		4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	
IIL2		53																																
IIL3		54																																
IIH1		55																																
IIH2		56																																
IIH3		57																																
IIH1	3006	58	GND	15 V	GND	15 V		15 V		15 V		15 V			GND	15 V	GND	15 V		15 V		15 V		15 V		15 V	QOA	14.95		14.95		4.95		Vdc
		59																									Q1A							
		60																									Q2A							

See footnotes at end of table.

MIL-M-38510/196A

Source: https://assist.dla.mil -- Downloaded on 2025-01-27T13:34:15Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,T,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Units
																												Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	RA	STA	ODA	DOA	QOA	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	ODB	DOB	QOB	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
V_{OH1}	3006	61	GND	15 V	GND	15 V		15 V		15 V		15 V		GND	GND	15 V	GND	15 V		15 V		15 V		15 V		15 V	03A	14.95		14.95		14.95		Vdc
		62																									00B							
		63																									01B							
		64																									02B							
		65																									03B							
V_{OL1}	3007	66				GND		GND		GND		GND						GND		GND		GND		GND			00A		0.05		0.05		0.05	
		67																									01A							
		68																									02A							
		69																									03A							
		70																									00B							
		71																									01B							
		72																									02B							
		73																									03B							
I_{IL}		74	5.0 V	GND [3]			0.4 V		0.4 V		0.4 V		0.4 V		5.0 V	GND			0.4 V		0.4 V		0.4 V		0.4 V	5.0 V	00A	0.51		0.36		0.64		mA
		75																									01A							
		76																									02A							
		77																									03A							
		78																									00B							
		79																									01B							
		80																									02B							
		81																									03B							

Footnotes at end of table.

MIL-M-38510/11A

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45:22
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,T,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Units
																												Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	RA	STA	ODA	QOA	Q0A	Q1A	Q1A	Q2A	Q2A	Q3A	Q3A	V_{SS}	RB	STB	Q0B	Q0B	Q0B	Q1B	Q1B	Q2B	Q2B	Q3B	Q3B	V_{DD}								
		Test No.																										Min	Max	Min	Max	Min	Max	
		82	15.0 V	GND	GND	GND	1.5 V	GND	1.5 V	GND	1.5 V	GND	1.5 V	GND	15.0 V	GND	GND	GND	1.5 V	GND	1.5 V	GND	1.5 V	GND	1.5 V	15.0 V	Q0A	3.4		2.4		4.2		mA
		83																									Q1A							
		84																									Q2A							
		85																									Q3A							
		86																									Q0B							
		87																									Q1B							
		88																									Q2B							
		89																									Q3B							
		90	GND	5.0 V		5.0 V	4.6 V	5.0 V	4.6 V	5.0 V	4.6 V	5.0 V	4.6 V		GND	5.0 V		5.0 V	4.6 V	5.0 V	4.6 V	5.0 V	4.6 V	5.0 V	4.6 V	5.0 V	Q0A	-0.51		-0.36		-0.64		
		91																									Q1A							
		92																									Q2A							
		93																									Q3A							
		94																									Q0B							
		95																									Q1B							
		96																									Q2B							
		97																									Q3B							
		98		15.0 V		15.0 V	13.5 V	15.0 V	13.5 V	15.0 V	13.5 V	15.0 V	13.5 V			15.0 V		15.0 V	13.5 V	15.0 V	13.5 V	15.0 V	13.5 V	15.0 V	13.5 V	15.0 V	Q0A	-3.4		-2.4		-4.2		
		99																									Q1A							
		100																									Q2A							
		101																									Q3A							
		102																									Q0B							

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2020-10-27T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	MIL-STD-883 method	Cases J,K,T,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Units
																												Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	RA	STA	ODA	DOA	QOA	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	QDB	DOB	QOB	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
		103	GND	15.0 V	GND	15.0 V		15.0 V		15.0 V		15.0 V		GND	GND	15.0 V	GND	15.0 V		15.0 V	13.5 V	15.0 V		15.0 V		15.0 V	Q1B	-3.4		-2.4		-4.2		mA
		104																					13.5 V				Q2B							
		105																							13.5 V		Q3B							
	3010	106	18.0 V	18.0 V	18.0 V	18.0 V		18.0 V		18.0 V		18.0 V			18.0 V	18.0 V	18.0 V	18.0 V		18.0 V		18.0 V		18.0 V		18.0 V	All inputs together		14					mA
		107	18.0 V	GND	GND	GND		GND		GND					GND	GND	GND	GND		GND		GND		GND			RA		1		45			
		108	GND	18.0 V	GND	GND																					STA							
		109		GND	18.0 V	GND																					ODA							
		110			GND	18.0 V																					DOA							
		111				GND		18.0 V																			D1A							
		112						GND		18.0 V																	D2A							
		113								GND																	03A							
		114										GND			18.0 V												RB							
		115													GND	18.0 V											STB							
		116														GND	18.0 V										DOB							
		117															GND	18.0 V									QOB							
		118																GND		18.0 V							D1B							
		119																GND		GND		18.0 V					D2B							
		120																GND		GND		GND		18.0 V			Q3B							
	3009	121																GND		GND		GND		GND			All inputs together		-14					

See footnotes at end of table.

MIL-M-38510/176A

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:43:42Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,L,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Units
		Symbol	RA	STA	3DA	DOA	Q0A	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	ST0	D0B	D0B	Q0B	C1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.																										Min	Max	Min	Max	Min	Max	
I_{IL2}	3009	122	GND	18.0 V	18.0 V	18.0 V		18.0 V		18.0 V		18.0 V		GND	18.0 V	18.0 V	18.0 V	18.0 V		18.0 V		18.0 V		18.0 V		18.0 V	RA		-1		-45			nA
		123	18.0 V	GND	18.0 V	18.0 V																					STA							
		124		18.0 V	GND	18.0 V																					ODA							
		125			18.0 V	GND																					D0A							
		126				18.0 V		GND																			D1A							
		127						18.0 V																			D2A							
		128								GND																	D3A							
		129								18.0 V																	RB							
		130										GND			GND												ST0							
		131										18.0 V			18.0 V												D0B							
		132															GND										D0B							
		133															18.0 V										D1B							
		134																									D2B							
		135																									D3B							
																												Subgroup 4 T_A = 25°C						
C_I	3012	136	6/																							GND	RA		7.5					pF
		137		6/																							STA							
		138			6/																						ODA							
		139				6/																					D0A							
		140						6/																			D1A							

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21 13:46Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,L,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits: Subgroup 8 T_A = 25°C Min	Subgroup 8 T_A = 25°C Max	Subgroup 7 T_A = -55°C Min	Subgroup 7 T_A = -55°C Max	Units
		Test no.	BA	STA	D3A	D0A	Q0A	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RD	STB	D0B	D0B	D0B	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}						
	3012	141	GND	5 V	GND	5 V	5 V	5 V	H	5 V	H	5 V	H	GND	5 V	5 V	GND	5 V	H	5 V	H	5 V	H	5 V	H	5 V	D2A		7.5			pF
		142	GND	GND	GND	5 V	5 V	5 V	L	5 V	L	5 V	L	GND	GND	GND	GND	5 V	L	GND	L	GND	L	GND	L	GND	D3A					
		143	GND	GND	GND	5 V	5 V	5 V	L	5 V	L	5 V	L	GND	GND	GND	GND	5 V	L	GND	L	GND	L	GND	L	GND	PB					
		144	GND	5 V	GND	5 V	5 V	5 V	H	5 V	H	5 V	H	GND	5 V	GND	GND	5 V	H	5 V	H	5 V	H	5 V	H	5 V	STB					
		145	GND	GND	GND	5 V	5 V	5 V	L	5 V	L	5 V	L	GND	GND	GND	GND	5 V	L	GND	L	GND	L	GND	L	GND	D0B					
		146	GND	5 V	GND	5 V	5 V	5 V	H	5 V	H	5 V	H	GND	5 V	GND	GND	5 V	H	5 V	H	5 V	H	5 V	H	5 V	D0B					
		147	GND	GND	GND	5 V	5 V	5 V	L	5 V	L	5 V	L	GND	GND	GND	GND	5 V	L	GND	L	GND	L	GND	L	GND	D0B					
		148	GND	5 V	GND	5 V	5 V	5 V	H	5 V	H	5 V	H	GND	5 V	GND	GND	5 V	H	5 V	H	5 V	H	5 V	H	5 V	D1B					
		149	GND	GND	GND	5 V	5 V	5 V	L	5 V	L	5 V	L	GND	GND	GND	GND	5 V	L	GND	L	GND	L	GND	L	GND	D2A					
		150	GND	5 V	GND	5 V	L	GND	L	GND	L	GND	L	GND	GND	5 V	GND	GND	L	GND	L	GND	L	GND	L	5 V	All outputs					
		151	GND	GND	GND	5 V	L	GND	L	GND	L	GND	L	GND	GND	GND	GND	GND	L	GND	L	GND	L	GND	L	GND						
		152	GND	GND	GND	5 V	L	5 V	L	5 V	L	5 V	L	GND	GND	GND	GND	5 V	L	GND	L	5 V	L	5 V	L	GND						
		153	GND	5 V	GND	5 V	H	5 V	H	5 V	H	5 V	H	GND	5 V	GND	GND	5 V	H	5 V	H	5 V	H	5 V	H	GND						
		154	GND	GND	GND	5 V	H	5 V	H	5 V	H	5 V	H	GND	5 V	GND	GND	5 V	H	5 V	H	GND	H	5 V	H	GND						
		155	GND	5 V	GND	GND	H	GND	H	GND	H	GND	H	GND	5 V	GND	GND	GND	H	GND	H	GND	H	5 V	L	5 V						
		156	GND	GND	GND	5 V	L	5 V	L	5 V	L	5 V	L	GND	GND	GND	GND	5 V	L	5 V	L	5 V	L	5 V	L	5 V						
		157	GND	5 V	GND	5 V	L	5 V	L	5 V	L	5 V	L	GND	GND	5 V	GND	5 V	L	5 V	L	5 V	L	5 V	L	5 V						

See 4/, 9/

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:45:2
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,T,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits					
																												Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C	
		Symbol	RA	STA	ODA	DOA	QOA	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	OEB	DOB	OEB	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}							
		Test No.																										Min	Max	Min	Max	Min	Max
t_{PHL1}	3003 Fig. 6	179	GND	IN1	GND									GND	GND	IN1	GND			IN2	OUT					5.0 V	STB to Q1B	17	335	23	470	17	335
		180																				IN2	OUT				STB to Q2B						
		181																						IN2	OUT		STB to Q3B						
t_{PLH1}		182				IN2	OUT																				STA to Q0A						
		183						IN2	OUT																		STA to Q1A						
		184								IN2	OUT																STA to Q2A						
		185										IN2	OUT														STA to Q3A						
		186																IN2	OUT								STB to Q0B						
		187																		IN2	OUT						STB to Q1B						
		188																				IN2	OUT				STB to Q2B						
		189																						IN2	OUT		STB to Q3B						
t_{PHL2}		190		5.0 V		IN2	OUT									5.0 V											D0A to Q0A	14	285	20	400	14	285
		191						IN2	OUT																		D1A to Q1A						
		192								IN2	OUT																D2A to Q2A						
		193										IN2	OUT														D3A to Q3A						
		194																IN2	OUT								D0B to Q0B						
		195																		IN2	OUT						D1B to Q1B						
		196																				IN2	OUT				D2B to Q2B						
		197																						IN2	OUT		D3B to Q3B						
t_{PLH2}		198				IN2	OUT																				D0A to Q0A						

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,R,T,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Unit
																												Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	0A	STA	0DA	0DA	Q0A	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	0DB	D0B	Q0B	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
t_{TLH}	3004 fig. 6	158	GND	5.0 V	GND	IN	OUT							GND	GND	5.0 V	GND									5.0 V	Q0A	14	290	14	280	14	280	ns
		159						IN	OUT																		Q1A							
		160								IN	OUT																Q2A							
		161										IN	OUT														Q3A							
		162																IN	OUT								Q0B							
		163																		IN	OUT						Q1B							
		164																				IN	OUT				Q2B							
		165																						IN	OUT		Q3B							
t_{THL}		166				IN	OUT																				Q0A							
		167						IN	OUT																		Q1A							
		168								IN	OUT																Q2A							
		169										IN	OUT														Q3A							
		170																IN	OUT								Q0B							
		171																		IN	OUT						Q1B							
		172																				IN	OUT				Q2B							
		173																						IN	OUT		Q3B							
t_{PHL1}	3003 fig. 6	174		IN1		IN2	OUT									IN1											STA to Q0A	17	335	23	470	17	335	
		175						IN2	OUT																		STA to Q1A							
		176								IN2	OUT																STA to Q2A							
		177										IN2	OUT														STA to Q3A							
		178																IN2	OUT								STB to Q0B							

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10-21T19:33:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,T,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Units
																												Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	RA	STA	CDA	DOA	QOA	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	DOB	DOB	QOB	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
t_{PLH2}	3003 Fig. 6	199	GND	5.0 V	GND			IN2	OUT					GND	GND	-5.0 V	GND									5.0 V	D1A to Q1A	14	285	20	400	14	285	ns
		200								IN2	OUT																D2A to Q2A							
		201										IN2	OUT														D3A to Q3A							
		202																IN2	OUT								D0B to Q0B							
		203																		IN2	OUT						D1B to Q1B							
		204																				IN2	OUT				D2B to Q2B							
		205																						IN2	OUT		D3B to Q3B							
t_{PHL3}		206	IN1			5.0 V	OUT	5.0 V		5.0 V		5.0 V			IN1			5.0 V		5.0 V		5.0 V		5.0 V			RA to Q0A	13	270	19	378	13	270	
		207							OUT																		RA to Q1A							
		208									OUT																RA to Q2A							
		209											OUT														RA to Q3A							
		210																	OUT								RB to Q0B							
		211																			OUT						RB to Q1B							
		212																					OUT				RB to Q2B							
		213																							OUT		RB to Q3B							
t_{PLH3}		214					OUT																				RA to Q0A							
		215							OUT																		RA to Q1A							
		216									OUT																RA to Q2A							
		217											OUT														RA to Q3A							
		218																	OUT								RB to Q3B							

See footnotes at end of table.

MIL-M-38510/74A

Source: https://assist.dla.mil -- Downloaded: 2020-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

MIL-STD-883 method	Cases J,K,T,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits: Subgroup 9 T_A = 25°C Min	Subgroup 9 T_A = 25°C Max	Subgroup 10 T_A = 125°C Min	Subgroup 10 T_A = 125°C Max	Subgroup 11 T_A = -55°C Min	Subgroup 11 T_A = -55°C Max	Units
	Symbol	RA	STA	CDA	2DA	QOA	D1A	Q1A	Q2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	CDB	D0B	Q0B	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}								
3003 Fig. 6	Test No.																																
	219	INI	5.0 V	GND	5.0 V		5.0 V				5.0 V		GND	INI	5.0 V	GND	5.0 V		5.0 V	OUT	5.0 V	OUT	5.0 V	OUT		RB to Q1B	10	200	14	270	10	280	ns
	220	GND		IN		OUT		OUT		OUT		OUT		GND		IN		OUT		OUT		OUT		OUT		RB to Q2B							
	221																									RB to Q3B							
	222																									QOA to Q0A							
	223																									QOA to Q1A							
	224																									QOA to Q2A							
	225																									QOA to Q3A							
	226																									QOB to Q0B							
	227																									QOB to Q1B							
	228																									QOB to Q2B							
	229																									QOB to Q3B							
	230																									QOA to Q0A							
	231																									QOA to Q1A							
	232																									QOA to Q2A							
	233																									QOA to Q3A							
	234																									QOB to Q0B							
	235																									QOB to Q1B							
	236																									QOB to Q2B							
	237																									QOB to Q3B							
	238																									QOA to Q0A							

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,T,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Units
		Symbol	RA	STA	ODA	DOA	QDA	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	QOB	DOB	QOB	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test No.																										Min	Max	Min	Max	Min	Max	
t_{PHZ}	See Fig. 6	239	GND	5.0 V	IN	5.0 V		5.0 V	OUT	5.0 V	OUT	5.0 V	OUT	GND	GND	5.0 V	GND	5.0 V	OUT	5.0 V	OUT	5.0 V	OUT	5.0 V	OUT	5.0 V	QOA to Q1A	11	220	16	310	11	220	ns
		240																									QOA to Q2A							
		241																									QOA to Q3A							
		242			GND												IN										QOB to QOB							
		243																									QOB to Q1B							
		244																									QOB to Q2B							
		245																									QOB to Q3B							
t_{PLZ}		246			IN	GND	OUT	GND	OUT	GND	OUT	GND	OUT				GND	GND	OUT	GND	OUT	GND	OUT	GND	OUT		QOA to QOA	9	180	13	255	9	180	
		247																									QOA to Q1A							
		248																									QOA to Q2A							
		249			GND												IN										QOA to Q3A							
		250																									QOB to QOB							
		251																									QOB to Q1B							
		252																									QOB to Q2B							
		253																									QOB to Q3B							
t_{PR} 10/		254	IN			5.0 V	OUT	5.0 V	OUT	5.0 V	OUT	5.0 V	OUT				GND	5.0 V		5.0 V		5.0 V		5.0 V			RA to QOA		200		280		200	
		255																									RA to Q1A							
		256																									RA to Q2A							
		257																									RA to Q3A							
		258	GND												IN				OUT								RB to QOB							

See footnotes at end of table.

MIL-M-38510/124

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,T,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Units
																												Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	RA	STA	DOA	BOA	QOA	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	DOB	D0B	Q0B	D1B	Q1B	D2B	Q2B	B3B	Q3B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
	See Fig. 6	259	GND	5.0 V	GND	5.0 V		5.0 V		5.0 V		5.0 V		GND	IN	5.0 V	GND	5.0 V		5.0 V	OUT	5.0 V	OUT	5.0 V		5.0 V	RB to Q1B		200		280		200	ns
		260																									RB to Q2B							
		261																							OUT		RB to Q3B							
PHL		262		IN1		IN2	OUT	IN2		IN2		IN2			GND	GND		IN2		IN2		IN2		IN2			STA to Q0A		140		200		140	
PLH		263							OUT																		STA to Q1A							
		264									OUT																STA to Q2A							
		265											OUT			IN1											STA to Q3A							
		266		GND															OUT								STB to Q0B							
		267																			OUT						STB to Q1B							
		268																					OUT				STB to Q2B							
		269																							OUT		STB to Q3B							
PHL		270		IN1		IN2	OUT									GND											D0A to STA		50		70		50	
PLH		271						IN2	OUT																		D1A to STA							
		272								IN2	OUT																D2A to STA							
		273										IN2	OUT														D3A to STA							
		274		GND												IN1		IN2	OUT		IN2	OUT					D0B to STB							
		275																									D1B to STB							
		276																				IN2	OUT				D2B to STB							
		277																						IN2	OUT		D3B to STB							
PHL		278		IN1		IN2	OUT									GND											STA to D0A		0		0		0	
PLH																																		

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:35:52Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case J,K,F,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Units
																												Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	RA	STA	ODA	D0A	Q0A	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	D0B	D1B	Q0B	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
t_{WHL} [10/]	See Fig. 6	279	GND	III	GND			IN2	OUT	IN2	OUT	IN2	OUT	GND	GND	GND	GND	IN2	OUT	IN2	OUT	IN2	OWT	IN2	OUT	5.0 V	STA to D1A		0		0		0	ns
		280																									STA to D2A							
		281																									STA to D3A							
		282		GND												IN1											STB to D0B							
		283																									STB to D1B							
		284																									STB to D2B							
		285																									STB to D3B							
																												Subgroup 12 T_A = 25°C						
t_{TLH1} and t_{THL}	3004 Fig. 6	286	GND	10 V	GND	IN	OUT	IN	OUT	IN	OUT	IN	OUT	GND	GND	10 V	GND	IN	OUT	IN	OUT	IN	OUT	IN	OUT	10.0 V	Q0A	5	100					ns
		287																									Q1A							
		288																									Q2A							
		289																									Q3A							
		290																									Q0B							
		291																									Q1B							
		292																									Q2B							
		293																									Q3B							
t_{PHL1} and t_{PLH1}	3003 Fig. 6	294		IN1		IN2	OUT	IN2	OUT	IN2	OUT	IN2	OUT			IN1											STA to Q0A	7	140					
		295																									STA to Q1A							
		296																									STA to Q2A							
		297																									STA to Q3A							

See footnotes at end of table.

MIL-M-38510/1961

Source: https://assist.dla.mil -- Downloaded: 2026/0-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	Fig. MIL-STD-883 method	Cases J,K,L,2 Symbol	Cases J,K,L,2 Test No.	1 BA	2 STA	3 D0A	4 Q0A	5 Q0A	6 D1A	7 Q1A	8 C2A	9 Q2A	10 D3A	11 Q3A	12 V_{SS}	13 RB	14 STB	15 D0B	16 D0B	17 Q0B	18 D1B	19 Q1B	20 D2B	21 Q2B	22 D3B	23 Q3B	24 V_{DD}	Measured terminal	Test limits Subgroup 12 T_A = 25°C Min	Max	Min	Max	Min	Max	Units
t_{PHL1} and t_{PLH1}	3003 Fig. 6		298	GND	IN1	GND	IN2								GND	GND	IN1		IN2	20*	IN2	OUT	IN2	OUT	IN2	OUT	10.0 V	STB to 00B	7	140				ns
			299																								STB to 01B						
			300																								STB to 02B						
			301																								STB to 03B						
t_{PHL2} and t_{PLH2}			302		10 V		IN2	OUT	IN2	OUT	IN2	OUT	IN2	OUT			10 V											D0A to 00B	6	120				
			303																									D1A to 01B						
			304																									D2A to 02B						
			305																									D3A to 03B						
			306																									D0B to 00B						
			307																									D1B to 01B						
			308																									D2B to 02B						
			309																									D3B to 03B						
t_{PHL3} and t_{PLH3}			310	IN1			10 V	OUT	10 V	OUT	10 V	OUT	10 V	OUT		IN1			10 V		10 V	OUT	10 V	OUT	10 V	OUT		BA to 00A	5	100				
			311																									BA to 01A						
			312																									BA to 02A						
			313																									BA to 03A						
			314																									RB to 00B						
			315																									RB to 01B						
			316																									RB to 02B						
			317																									RB to 03B						

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases 1,2,3,4,5	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Subgroup 12 T_A = 25°C Min	Max	Min	Max	Units
		Symbol	RA	STA	D2A	I0A	Q0A	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	CKB	D0B	Q0B	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}						
		Test No.																														
t_{PWZ} and t_{PZH}	See fig. 6	318	GND	10 V	IN	10 V	OUT	10 V	OUT	10 V	OUT	10 V	OUT	GND	GND	10 V	GND	10 V	OUT	10 V	OUT	10 V	OUT	10 V	OUT	10 V	Q0A to Q3A	6/5	120/ 100			ns
		319																									Q0A to Q1A					
		320																									Q0A to Q2A					
		321																									Q0A to Q3A					
		322																									Q0B to Q0B					
		323																									Q0B to Q1B					
		324																									Q0B to Q2B					
		325																									Q0B to Q3B					
t_{PWZ} and t_{PZL}		326	GND	10 V	IN	GND	OUT	GND	OUT	GND	OUT	GND	OUT	GND	GND	10 V	GND	GND	OUT	GND	OUT	GND	OUT	GND	OUT	10 V	Q0A to Q0A	5	100			ns
		327																									Q0A to Q1A					
		328																									Q0A to Q2A					
		329																									Q0A to Q3A					
		330																									Q0B to Q0B					
		331																									Q0B to Q1B					
		332																									Q0B to Q2B					
		333																									Q0B to Q3B					
t_{PR} LO/	Fig. 6	334	IN	10 V	IN	10 V	OUT	10 V	OUT	10 V	OUT	10 V	OUT	GND	GND	10 V	GND	10 V	OUT	10 V	OUT	10 V	OUT	10 V	OUT	10 V	RA to Q0A	140				ns
		335																									RA to Q1A					
		336																									RA to Q2A					
		337																									RA to Q3A					

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-21T13:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,E,T,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits						Units
																												Subgroup 12 T_A = 25°C						
		Symbol	RI	STA	ODA	DOA	QOA	D1A	Q1A	D2A	Q2A	D3A	Q3A	V_{SS}	RB	STB	OOB	DOB	QOB	D1B	Q1B	D2B	Q2B	D3B	Q3B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
t_{PLH} 1/	Fig. 6	338	GND	10 V	GND	10 V		10 V	OUT	10 V		10 V		GND	GND	10 V	GND	10 V	OUT	10 V		10 V		10 V		10.0 V	R0 to Q0B		140					ns
		339																			OUT						R1 to Q1B							
		340																					OUT				R2 to Q2B							
		341																							OUT		R3 to Q3B		10					
t_{PHL} 1/		342		IN1		IN2	OUT	IN2		IN2		IN2			GND	GND		IN2		IN2		IN2		IN2			STA to Q0A		80					
		343							OUT																		STA to Q1A							
		344									OUT																STA to Q2A							
		345											OUT														STA to Q3A							
		346		GND												IN1			OUT								STB to Q0B							
		347																			OUT						STB to Q1B							
		348																					OUT				STB to Q2B							
		349																							OUT		STB to Q3B		10					
t_{SLH} 1/		350		IN1			OUT									GND											D0A to S1A		30					
		351						IN2	OUT																		D1A to S1A							
		352								IN2	OUT																D2A to S1A							
		353										IN2	OUT														D3A to S1A							
		354		GND												IN1		IN2	OUT								D0B to S1B							
		355																		IN2	OUT						D1B to S1B							
		356																				IN2	OUT				D2B to S1B							
		357																						IN2	OUT		D3B to S1B		10					

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21 13:45:52
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases 1,3,4,7 Symbol	Test No.	1 RB	2 STA	3 DOA	4 D0A	5 D0A	6 D1A	7 Q1A	8 D2A	9 Q2A	10 D3A	11 Q3A	12 V_{SS}	13 RB	14 STB	15 D0B	16 D0B	17 Q0B	18 D1B	19 Q1B	20 D2B	21 Q2B	22 D3B	23 Q3B	24 V_{DD}	Measured terminal	Test limits Subgroup 12 T_A = 25°C Min	Max	Test limits Min	Max	Test limits Min	Max	Units
t_{PHL}, t_{PLH}	Fig. 6	358		GND	I#1	GND	I#2	OUT	I#2	OUT	I#2	OUT	I#2	OUT	GND	GND	GND	I#2	OUT	I#2	OUT	I#2	OUT	I#2	OUT	10.0 V	STA to D0A	0					ns
		359																									STA to D1A						
		363																									STA to D2A						
		361																									STA to D3A						
		362																									STB to D0B						
		383																									STB to D1B						
		364																									STB to D2B						
		365																									STB to D3B						

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

1/ Pins not designated may be "High" level logic, "Low" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ Clock device to time slot 26 of sequence B in 9/ before performing leakage test.

4/ The input/output conditions, contained in 9/ shall be used. For device type 01, the timing sequence specified in 9/, sequence B, (except for the truth table test) shall be used; for device type 02, the timing sequence specified in 9/, sequence A shall be used.

5/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

6/ See 4.4.1c.

7/ The truth table shall be performed using the input/output conditions and timing sequence contained in 9/, sequence C.

8/ Write disable (WD) must be held at logic "1" while changing the address lines (A0, A1, A2) to select the next output bit.

9/ The following timing sequence and input/output conditions shall apply:

Input/output conditions

Test	V_{DD}	Input levels		Output levels	
		1	0	1	0
	Volts	Volts	Volts	Min	Max
V_{IL1} V_{IH1}	5.0	V_{DD} 3.5	1.5 V_{SS}	4.5 V	0.5 V
V_{IL2} V_{IH2}	10.0	V_{DD} 7.0	3.0 V_{SS}	9.0 V	1.0 V
V_{IL3} V_{IH3}	15.0	V_{DD} 11.0	4.0 V_{SS}	13.5 V	1.5 V
Truth table	≤5.0 ≥18.0	V_{DD}	V_{SS}	V_{DD} -0.5 V	V_{SS} +0.5 V

Sequence A

	Time slot	1	2	3	4	5	6	7	8
INPUTS	RA	0	0	0	0	0	0	1	0
	STA	1	0	0	1	0	0	0	0
	ODA	0	0	0	0	0	0	0	0
	DOA	0	0	1	1	1	0	1	1
	D1A	0	0	1	1	1	0	1	1
	D2A	0	0	1	1	1	0	1	1
	D3A	0	0	1	1	1	0	1	1
	RB	0	0	0	0	0	0	1	0

Source: https://assist.dla.mil -- Downloaded: 20260-21T13:45Z
Check the source to verify that this is the current version before use.

Sequence A (continued)

	Time slot	1	2	3	4	5	6	7	8
INPUTS	STB	1	0	0	1	0	0	0	0
	ODB	0	0	0	0	0	0	0	0
	D0B	0	0	1	1	1	0	1	1
	D1B	0	0	1	1	1	0	1	1
	D2B	0	0	1	1	1	0	1	1
	D3B	0	0	1	1	1	0	1	1
OUTPUTS	Q0A	0	0	0	1	1	1	0	0
	Q1A	0	0	0	1	1	1	0	0
	Q2A	0	0	0	1	1	1	0	0
	Q3A	0	0	0	1	1	1	0	0
	Q0B	0	0	0	1	1	1	0	0
	Q1B	0	0	0	1	1	1	0	0
	Q2B	0	0	0	1	1	1	0	0
	Q3B	0	0	0	1	1	1	0	0

Sequence B

	Time slot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29
INPUTS	A2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0
	A1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0
	A0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	0	0	0	1	1	1	0	0	0	1	1	1	0	0	0
	R	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
	D	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	WD	1	1	1	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1	1	1
OUTPUTS	Q0	X	0	0	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	0
	Q1	X	0	0	X	0	0	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	0
	Q2	X	0	0	X	0	0	X	0	0	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	0
	Q3	X	0	0	X	0	0	X	0	0	X	0	0	X	1	1	X	1	1	X	1	1	X	1	1	X	1	1	X	0
	Q4	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	1	1	X	1	1	X	1	1	X	1	1	X	0
	Q5	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	1	1	X	1	1	X	1	1	X	0
	Q6	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	1	1	X	1	1	X	0
	Q7	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	0	0	X	1	1	X	0

Source: https://assist.dla.mil - Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

SEQUENCE C

	Time slot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
INPUTS	A2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	A1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	A0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	R	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	
	D	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	0	0	
	WD	1	1	1	0	1	1	0	1	1	0	1	1	1	0	0	0	0	
OUTPUTS	Q0	X	0	0	X	1	1	X	0	0	X	1	X	0	X	1	X	0	BIT 0
	Q1	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q2	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q3	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q4	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q5	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q6	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q7	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	

	Time slot	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	
INPUTS	A2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	A1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	A0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	R	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	
	D	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	0	0	
	WD	1	1	1	0	1	1	0	1	1	0	1	1	1	0	0	0	0	
OUTPUTS	Q0	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	BIT 1
	Q1	X	0	0	X	1	1	X	0	0	X	1	X	0	X	1	X	0	
	Q2	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q3	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q4	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q5	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q6	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q7 P1	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

SEQUENCE C - Continued

	Time slot	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	
INPUTS	A2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	A1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	A0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	R	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	
	D	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	0	0	
	WD	1	1	1	0	1	1	0	1	1	0	1	1	1	0	0	0	0	
OUTPUTS	Q0	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	BIT 2
	Q1	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q2	X	0	0	X	1	1	X	0	0	X	1	X	0	X	1	X	0	
	Q3	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q4	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q5	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q6	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q7	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	

	Time slot	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	
INPUTS	A2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	A1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	A0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	R	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	
	D	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	0	0	
	WD	1	1	1	0	1	1	0	1	1	0	1	1	1	0	0	0	0	
OUTPUTS	Q0	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	BIT 3
	Q1	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q2	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q3	X	0	0	X	1	1	X	0	0	X	1	X	0	X	1	X	0	
	Q4	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q5	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q6	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q7	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	

Source: https://assist.dla.mil - Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

SEQUENCE C - Continued

	Time slot	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	
INPUTS	A2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	A1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	A0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	R	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	
	D	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	0	0	
	WD	1	1	1	0	1	1	0	1	1	0	1	1	1	0	0	0	0	
OUTPUTS	Q0	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	BIT 4
	Q1	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q2	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q3	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q4	X	0	0	X	1	1	X	0	0	X	1	X	0	X	1	X	0	
	Q5	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q6	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q7	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	

	Time slot	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	
INPUTS	A2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	A1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	A0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	R	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	
	D	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	0	0	
	WD	1	1	1	0	1	1	0	1	1	0	1	1	1	0	0	0	0	
OUTPUTS	Q0	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	BIT 5
	Q1	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q2	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q3	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q4	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q5	X	0	0	X	1	1	X	0	0	X	1	X	0	X	1	X	0	
	Q6	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q7	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

SEQUENCE C - Continued

	Time slot	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	
INPUTS	A2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	A1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	A0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	R	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	
	D	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	0	0	
	WD	1	1	1	0	1	1	0	1	1	0	1	1	1	0	0	0	0	
OUTPUTS	Q0	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	BIT 6
	Q1	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q2	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q3	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q4	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q5	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q6	X	0	0	X	1	1	X	0	0	X	1	X	0	X	1	X	0	
	Q7	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	

	Time slot	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	
INPUTS	A2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	A1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	A0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
	R	1	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	
	D	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	0	0	
	WD	1	1	1	0	1	1	0	1	1	0	1	1	1	0	0	0	0	
OUTPUTS	Q0	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	BIT 7
	Q1	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q2	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q3	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q4	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q5	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q6	X	0	0	X	0	0	X	0	0	X	0	X	0	X	0	X	0	
	Q7	X	0	0	X	1	1	X	0	0	X	1	X	0	X	1	X	0	

10/ The requirements for t_{PR}, t_{PS}, t_{SLH}, and t_{HHL} are considered met by setting the parameter to the values given in the limits column and observing proper output state changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

c. End-point electrical parameters shall be as specified in table II herein and shall consist only of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 herein and as shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit V_{SS} terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25°C ±3°C; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 25°C.

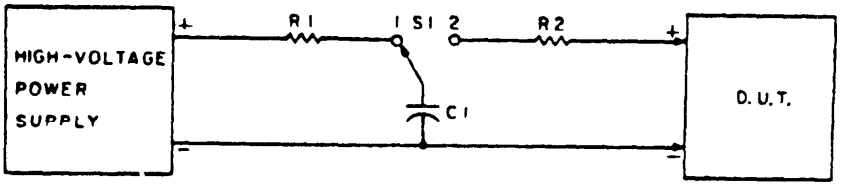
Parameter 1/	Device types	
	01	02
I_{SS}	±10 nA	±10 nA
I_{OL1}	±15%	±15%
I_{OH1}	±15%	±15%

1/ Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of 4) of the devices under test (DUT) shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit shown on figure 7.

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD}. Measure I_{SS} on the DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at V_{DD} = 18 Vdc.



V_{ZAP} = 400 V charge on C1.
1 MΩ ≤ R1 ≤ 50 MΩ
R2 = 1.5 kΩ
C1 = 100 pF
S1 = Hg-wetter "bounceless" relay.

FIGURE 7. High voltage (V_{ZAP}) test circuit.

b. V_{ZAP} is applied to DUT in the following modes (see table V) by changing C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated output

c. Within 24 hours repeat the I_{SS}, I_{IL}, and I_{IH} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS}) test. When performing quiescent supply current measurements, the meter shall be placed so that all currents flow through the meter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and as specified herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the RHA level being tested, and meet the end-point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end-point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table VIII and on figure 8. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardening end-point electrical parameters at 25°C.

Parameter	All device types	V_{DD}
V_{TN}	0.3 V min	10 V
V_{TP}	2.8 V max	10 V
ΔV_T	1.4 V	10 V
I_{SS}	100 x max limit	15 V
t_{PLH}	1.35 x max limit	5 V
t_{PHL}	1.35 x max limit	5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	V_{DD} = 10 Vdc (through a 30- to 60-kΩ resistor)	V_{SS} = GND	V_{DD} = 10 Vdc
01	2, 3, 4, 5, 6, 7	8	16
02	1,2,4,6,8,10,13,14,16,18,20,22	3,12,15	24

Pins not designated are open or connected to 10 Vdc through a 30- to 60-kilohm resistor.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady state life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameters data (see 4.2d).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

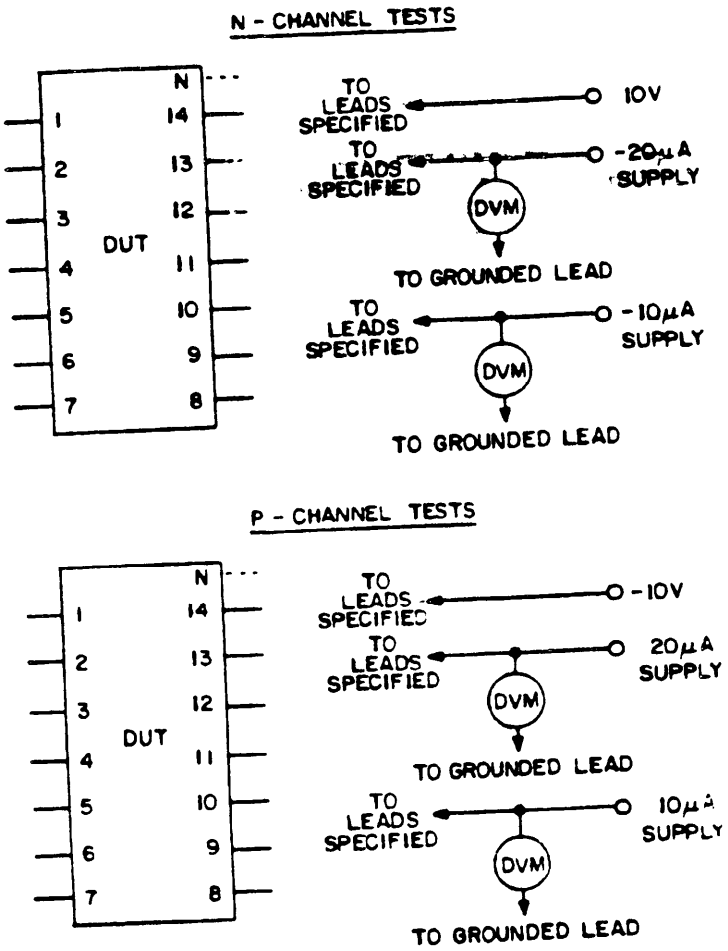


FIGURE 8. Threshold-voltage test circuit.

TABLE VIII. Threshold-voltage test circuit conditions.

Device type	GND	10V	V_{TN} measured at		GND	-10V	V_{TP} measured at	
			-20 μA supply	-10 μA supply			20 μA supply	10 μA supply
01	3	2,4,5,6,7,16		8	3	2,4,5,6,7,8		16
02	2	1,3,4,6,8,10 13,14,15,16,18 20,22,24		12	2	12		1,3,4,6,8,10 13,14,15,16,18 20,22,24

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging require
with MIL-M-38510.

6. NOTES

6.1 Intended use. Mi
original equipment desig

6.2 Ordering data. 1

a. Complete part nu

b. Requirements for
data pertinent t
by the device ma

c. Requirements for

d. Requirements for
contracting acti
if applicable.

e. Requirements for
method 5003 of M
applicable.

f. Requirements for

g. Requirements
applicable. The
otherwise specif
by, or direct st

h. Requirements for

i. Requirements for
applicable.

6.3 Abbreviations, sy
definitions used herein

C_I - - - - - - - - -
GND - - - - - - - - -
I_{SS} - - - - - - - - -
T_A - - - - - - - - -
V_{DD} - - - - - - - - -
V_{SS} - - - - - - - - -
V_{ZAP} - - - - - - - -

6.4 Logistic support.
Unless otherwise specif
will be acquired to dev
or class B for Departme
length leads and lead f

6.5 Substitutability
convenience of users.
replace the listed gene
not have equivalent ope
temperature ranges or m
may have slight phys
information shall no
for MIL-M-38510 types o

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

Military device type	Generic-industry type
01	4099B
02	4508B

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AR, SM
Navy - AS, CG, MC, OS, SH

Agent:
DLA - ES

Preparing activity:
NASA - NA

(Project 5962-0639-7)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

National Aeronautics and
Space Administration

Washington, D C
20546

Official Business
Penalty for Private Use $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12028 WASHINGTON, D C

POSTAGE WILL BE PAID BY NASA

George C. Marshall Space Flight Center
National Aeronautics and Space Administration
Attn: EG02
Marshall Space Flight Center, AL 35812

NASA

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/176A	MICROCIRCUITS, DIGITAL, CMOS, LATCHES, MONOLITHIC SILICON, POSITIVE LOGIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:45Z
Check the source to verify that this is the current version before use.